

Large language models in digital forensics: capabilities, challenges and future directions

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HIGHLIGHTS

- Systematic review of 33 works mapping LLM capabilities to DFRWS forensic phases.
- LLMs achieve 85%–98% accuracy across diverse forensic tasks with limitations.
- Probabilistic LLM outputs conflict with deterministic forensic requirements.
- Three strategic integration points identified for LLM-assisted investigations.

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ABSTRACT

The rapid advancement of large language models (LLMs) has simultaneously created opportunities and challenges for digital forensic science. This survey systematically examines the emerging intersection between generative artificial intelligence and digital forensics through our analysis of 33 peer-reviewed works. We map LLM capabilities across the established Digital Forensic Research Workshop (DFRWS) process model, identifying three strategic integration points where these technologies demonstrate measurable benefits – pattern recognition during the examination phase, evidence analysis during the analysis phase, and evidence presentation and reporting during the presentation phase. Our findings show that LLMs achieve substantial performance improvements across diverse forensic tasks, but critical challenges persist, including the fundamental tension between the probabilistic nature of LLM outputs and deterministic forensic requirements, alongside concerns regarding explainability, reproducibility, and legal admissibility. We identify significant research gaps in validation frameworks, forensic-ready architectures, and standardised evaluation protocols. The survey establishes a comprehensive research agenda spanning technical, methodological, and legal domains, emphasising the necessity for interdisciplinary collaboration and human-AI collaborative approaches to preserve forensic integrity when leveraging LLM capabilities.

1. Introduction

The field of digital forensics has continuously evolved to address the challenges posed by emerging technologies, from the proliferation of personal computers and mobile devices to the advent of cloud computing and the Internet of Things (IoT). Each technological shift has necessitated the development of new investigative methodologies, tools, and frameworks to effectively identify, preserve, collect, examine, and analyse digital evidence. The most recent transformation in this landscape stems from the rapid advancement and widespread adoption of generative artificial intelligence (AI), particularly large language models (LLMs) such as ChatGPT, GPT-4, Claude, and Gemini.

These sophisticated AI systems, trained on vast corpora of text data, demonstrate remarkable capabilities in natural language understanding, pattern recognition, and automated reasoning that closely align with the core analytical requirements in digital forensic investigations (Scanlon et al., 2023a). The emergence of LLMs presents an opportunity for the forensic community to explore how these powerful AI capabilities can enhance traditional forensic workflows to address critical scalability challenges facing contemporary digital forensic science (Du et al., 2020; Casino et al., 2022).

However, the integration of LLMs into digital forensic practice introduces fundamental tensions between the capabilities these models offer

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and the rigour of digital forensic science. Existing digital forensic tools and methodologies, which have been designed for deterministic systems with predictable evidence patterns, face significant challenges when confronted with probabilistic outputs and non-deterministic behaviour of LLM-based systems (Wickramasekara and Scanlon, 2024). The foundational principles of digital forensics, specifically reproducibility, chain of custody maintenance, and legal admissibility, must be carefully reconsidered when incorporating AI-assisted capabilities that may generate varying outputs across multiple invocations of identical inputs such as those generated by LLMs. This convergence of opportunities and challenges establishes a critical research gap at the intersection of AI systems and digital forensic science, where existing methodologies must evolve to accommodate the application of LLM capabilities to enhance forensic workflows.

This survey provides a comprehensive examination of the emerging intersection between generative AI, specifically LLMs, and digital forensics, synthesising the rapidly evolving body of research that has emerged since the mainstream adoption of ChatGPT in late 2022. Based on a systematic literature review of 33 peer-reviewed works published between 2023 and 2025, we map the current landscape of LLM applications across the established Digital Forensic Research Workshop (DFRWS) process model (Palmer, 2001), identifying both promising capabilities and critical limitations.

Our analysis demonstrates a clear progression from exploratory investigations, to methodological developments and growing empirical validation efforts, with research efforts predominantly concentrated in the examination and analysis phases of digital forensic investigations. We identify significant gaps in current research such as limited attention to validation controls necessary for maintaining forensic integrity when using AI-assisted methods, and insufficient exploration of automated structured evidence representation using established ontological frameworks. This survey makes three primary contributions:

1. A systematic mapping of LLM capabilities to forensic processes based on the foundational DFRWS process model, identifying integration points where these technologies can provide measurable benefits whilst highlighting areas where human expertise remains essential.
2. Synthesis of key challenges that must be addressed for the practical deployment of LLM-assisted forensic methods in operational digital forensic contexts.
3. A comprehensive research agenda that identifies priority areas for future investigation, including the development of forensic-ready LLM architectures, standardised validation frameworks, and interdisciplinary approaches.

The remainder of this survey is organised as follows. Section 2 establishes the foundational concepts necessary for understanding the intersection of LLMs and digital forensics, including the DFRWS process model, LLM architectures and capabilities, and the unique characteristics of LLM-generated evidence. Section 3 describes our systematic literature review methodology encompassing the search strategy and the associated inclusion criteria, as well as the analytical approach used to identify and synthesise relevant research based on published state-of-the-art. Section 4 examines LLM-assisted forensic capabilities across the DFRWS process model, highlighting demonstrated capabilities and their inherent limitations through systematic mapping of the included literature. Section 5 provides a comprehensive analysis of the technical, methodological, and other challenges facing LLM-assisted digital forensics, including issues of explainability, reproducibility, and forensic integrity. Section 6 outlines future research directions across technical, methodological, and legal domains, establishing a comprehensive research agenda for advancing the intersection of generative AI and digital forensics. Finally, Section 7 discusses the implications of our findings for both research and practice, and Section 8 concludes with a synthesis of

key contributions and a call to action for interdisciplinary collaboration in this rapidly evolving domain.

2. Background and terminology

This section introduces the core concepts, terminologies, and frameworks that underpin subsequent discussions on LLM integration into forensic workflows.

2.1. Digital forensics fundamentals

Digital forensics encompasses the scientifically derived and validated methods used to identify, preserve, collect, analyse, interpret, document, and present digital evidence derived from digital sources (Palmer, 2001). The field has evolved significantly from its traditional origins in computer forensics to encompass diverse technological domains including mobile devices, cloud computing, and Internet of Things (IoT) systems (Al-Dhaqm et al., 2021). Understanding the foundational principles and processes of digital forensics is essential for examining how LLM capabilities can be integrated into investigative workflows alongside the required scientific rigour.

2.1.1. The DFRWS process model

The Digital Forensic Research Workshop (DFRWS) process model (Palmer, 2001), shown in Fig. 1, established the foundational framework for conducting digital investigations, encompassing six core phases that guide forensic practitioners through systematic evidence-handling workflows. The *identification* phase involves recognising potential sources of digital evidence across diverse electronic devices and systems, from traditional computers to contemporary IoT devices and cloud infrastructure. The *preservation* phase ensures the integrity of identified evidence through approaches such as cryptographic hashing and appropriate storage mechanisms, maintaining an unbroken chain of custody essential for subsequent evidential admissibility. The *collection* phase focuses on the acquisition of digital artefacts using forensically sound methods, typically involving write-blocking devices and validated forensic tools to create bit-for-bit copies of storage media. The *examination* phase entails systematic assessment and extraction of relevant information from collected data, often requiring specialised techniques to circumvent encryption, recover deleted files, and process large volumes of heterogeneous data. The *analysis* phase represents the interpretive element of digital forensic investigations, where practitioners may need to correlate evidence across multiple sources, develop and test hypotheses, and subsequently establish causal relationships

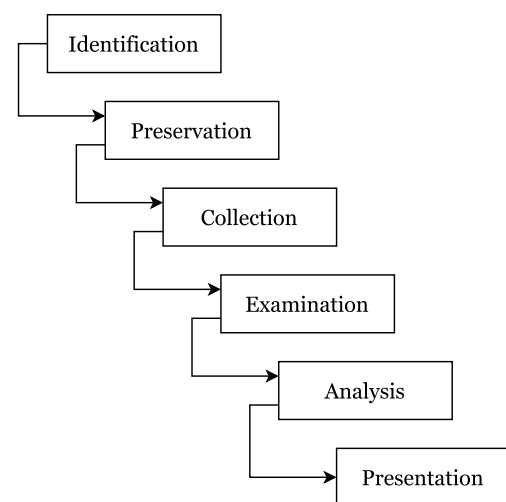


Fig. 1. The phases of the digital forensic research workshop (DFRWS) process model. Adapted from (Palmer, 2001).

between digital artefacts and investigative questions (Casey, 2011). Finally, the *presentation* phase involves documenting findings and providing expert testimony.

2.1.2. Evolution of digital evidence sources

Digital evidence sources have evolved from the early focus on personal computer file systems to encompass increasingly complex and distributed technological ecosystems. Traditional evidence sources including hard drives, memory dumps, and network logs have been supplemented by cloud-based storage, mobile device artefacts, and IoT sensor data (Khan et al., 2016). The emergence of LLMs and LLM agents represents the latest technological shift in this evolution, introducing fundamentally new categories of digital evidence that differ qualitatively from traditional artefacts. LLM-integrated systems generate evidence across multiple architectural layers, including conversational logs capturing natural language interactions, agent execution traces documenting autonomous decision-making processes, tool invocation records showing external API calls, and reasoning chains that capture intermediate steps in multi-stage processing (Wang et al., 2024). These evidence types also present unique challenges for traditional forensic methodologies, as they combine structured metadata with unstructured natural language content, require an understanding of conversational flow, and may span multiple system boundaries in distributed architectures.

2.2. Generative AI and large language models

This subsection examines the evolution from traditional AI approaches to contemporary LLMs, establishing the baseline foundation necessary for understanding both their forensic applications and the unique challenges they introduce.

2.2.1. From Artificial Intelligence to Generative AI

The field of AI encompasses computational systems designed to perform tasks that typically require human intelligence – such as reasoning, learning, and problem-solving. Within the broader domain of AI, machine learning (ML) represents a subset of AI techniques that enable such systems to learn patterns from data in the context of specific tasks, while deep learning, as a subset of ML, specialises in neural network architectures with multiple layers that can automatically extract hierarchical features from raw data (Brown et al., 2020). As shown in Fig. 2, generative AI represents a distinct category within deep learning, characterised by models capable of generating content including text, images, audio, and code based on learned patterns from training data. Unlike discriminative ML models that classify or predict based on input data, generative

models learn the underlying probability distributions of their training data to produce new instances resembling the original corpus (Shao et al., 2024).

LLMs constitute a prominent class of generative AI systems built upon transformer architectures (Vaswani et al., 2017). Transformer LLMs process and generate human-like text through self-attention mechanisms and are trained on vast corpora comprising billions of tokens. LLMs demonstrate emergent capabilities in natural language understanding, reasoning, and task generalisation that distinguish them from earlier natural language processing approaches. Thus, contemporary LLMs such as GPT, Claude, and Gemini families carry sophisticated abilities in contextual comprehension and multi-step reasoning – capabilities that align closely with the analytical requirements of digital forensic investigations (Scanlon et al., 2023a).

2.2.2. LLM architectures and forensically relevant capabilities

The transformer architecture behind modern LLMs relies on self-attention mechanisms that enable parallel processing of entire text sequences, thereby allowing models to capture long-range dependencies and contextual relationships required for understanding complex narratives (Vaswani et al., 2017). This architectural foundation supports several capabilities directly relevant to digital forensic applications:

1. *Contextual understanding* allows LLMs to interpret ambiguous references, resolve pronouns, and maintain coherence across extended conversational exchanges (Nair et al., 2020), which can prove valuable when analysing communication logs and reconstructing event sequences from fragmented evidence.
2. *Pattern recognition capabilities* allow LLMs to identify anomalous behaviours and attack signatures, as well as correlate disparate evidence sources without requiring explicit rule-based programming (Lin, 2024).
3. *Semantic reasoning* allows models to infer implicit relationships between entities and generate hypotheses about causal connections, alongside explanations and reasoning that could support investigative decision-making (Wickramasekara and Scanlon, 2024).

However, these capabilities come with inherent limitations that directly impact digital forensic applications. The probabilistic nature of LLM output generation, controlled by parameters such as temperature, introduces non-determinism where identical inputs may produce varying outputs across multiple invocations (Ouyang et al., 2022). This characteristic fundamentally conflicts with the reproducibility requirements essential for digital forensic method acceptance and the legal admissibility of digital evidence. Furthermore, LLMs demonstrate a strong susceptibility to hallucinations (Zhang et al., 2025), generating incredibly plausible, yet factually incorrect information that can mislead investigations if not properly validated. Finally, the context window limitations inherent to transformer architectures also potentially constrain the volume of information that can be processed simultaneously, with contemporary models ranging from thousands to millions of tokens, potentially fragmenting the analysis of large-scale forensic datasets (Liu et al., 2024).

3. Systematic literature review methodology

Having established the foundational concepts spanning digital forensics and LLM technologies, we now turn to examine the current state of research at this intersection.

3.1. Review protocol

This systematic literature review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2010) to develop comprehensive and reproducible coverage of research at the intersection of generative AI and digital forensics. The review protocol includes four key components – search

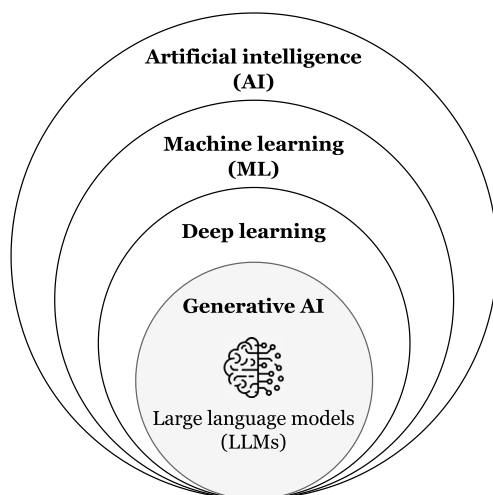


Fig. 2. The relationship between artificial intelligence (AI), machine learning (ML), deep learning, generative AI and large language models (LLMs).

strategy, inclusion and exclusion criteria, workflow adaptation, and quality assessment.

The *search strategy* employed ego graph analysis to identify relevant search terms across two conceptual domains. For generative AI, the ego graph centred on “generative AI” and produced terms including *LLM*, *large language model*, *GPT*, and *ChatGPT*. For digital forensics, the ego graph centred on “digital forensics” and yielded terms including *computer forensics* and *cyber forensics*. These terms were combined using Boolean operators to construct database-specific queries targeting four major research databases – Scopus, Web of Science, IEEE Xplore, and ScienceDirect. The search results were limited to items published by October 31, 2025, to capture the most recent developments in this rapidly evolving field.

Inclusion criteria required that records: (1) contain at least one search term from each conceptual domain in the title, abstract, or keywords; (2) be published between 2023 and 2025; (3) be associated with computer science, engineering, or social sciences subject areas where such filtering was supported by the database; (4) be written in English; and (5) represent peer-reviewed journal articles or conference papers. *Exclusion criteria* eliminated records lacking essential metadata (title, year, source, abstract, DOI), non-English publications, grey literature, and works not addressing LLM applications in digital forensics. As shown in Fig. 3, the initial database queries returned 248 records, which were reduced to 33 included works following deduplication, screening, and full-text eligibility assessment.

The PRISMA workflow adaptation involved three sequential filtering stages. First, automated deduplication removed identical records based on title, year, and DOI matching. Second, abstract screening applied the inclusion criteria to eliminate irrelevant works, with particular attention to identifying studies addressing LLM-assisted forensic analysis or closely related domains such as cyber incident response. Third, full-text eligibility assessment examined the remaining works to confirm substantive contributions to the intersection of generative AI and digital forensics, with particular focus on methodological rigour and empirical validation where applicable.

Finally, *quality assessment* employed natural language processing techniques, including stopword removal and Porter stemming (Willett, 2006), to facilitate automated analysis of abstracts and full texts. Each included work was systematically mapped to the DFRWS process model phases to assess coverage across the forensic workflow, with particular attention to methodological contributions, empirical validation status, and model architectures explored.

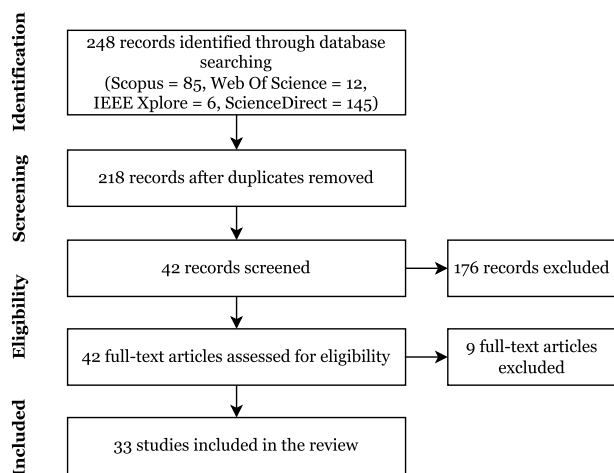


Fig. 3. PRISMA (Preferred reporting items for systematic reviews and meta-analyses) flowchart of study inclusions and exclusions for the systematic literature review.

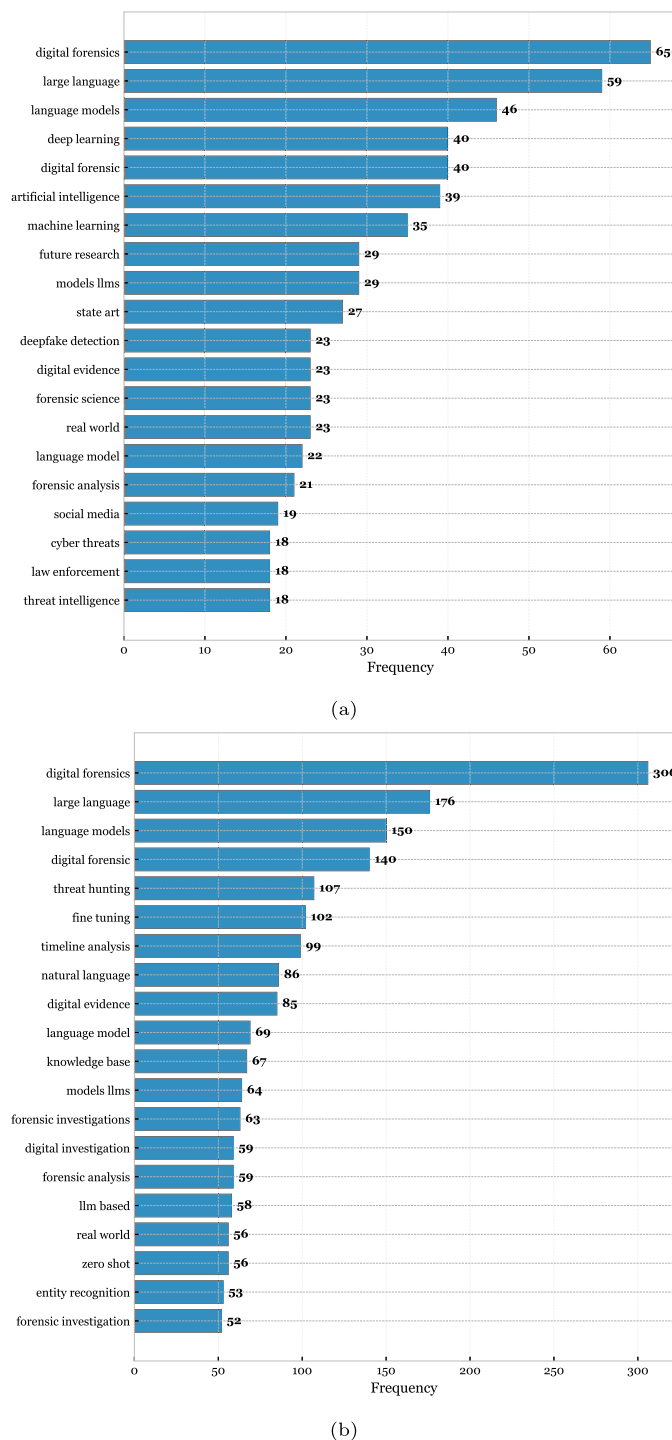


Fig. 4. Top 20 bigrams from abstract (a) and included full-text record (b) analysis.

3.2. Literature landscape analysis

Bigram frequency analysis of the included literature, as shown in Fig. 4, identifies the conceptual landscape spanning LLM-assisted digital forensics. Abstract analysis (Fig. 4(a)) demonstrates that *digital forensics* (65 occurrences) and *large language* (59 occurrences) represent the most prominent themes, followed by *language models* (46 occurrences), *deep learning* (40 occurrences), and *digital forensic* (40 occurrences). Notably, domain-specific applications emerge

through terms such as *deepfake detection* (23 occurrences), *threat intelligence* (18 occurrences), and *cyber threats* (18 occurrences), alongside *law enforcement* (18 occurrences), highlighting practical investigative contexts.

Full-text analysis (Fig. 4(b)) further shows substantially higher frequency counts, with *digital forensics* appearing 306 times, *large language* 176 times, and *language models* 150 times. The detailed examination also identifies specific methodological concepts, including *fine-tuning* (102 occurrences), *timeline analysis* (99 occurrences), *threat hunting* (107 occurrences), and *zero-shot* (56 occurrences), which demonstrate a diverse range of technical approaches explored across the literature. The prominence of *natural language* (86 occurrences) and *entity recognition* (53 occurrences) underscores the focus on natural language processing capabilities inherent to LLMs. Furthermore, *digital investigation* (59 occurrences) and *forensic investigations* (63 occurrences) collectively emphasise the investigative methodology focus, whereas *knowledge base* (67 occurrences) and *LLM-based* (58 occurrences) indicate growing attention to knowledge-driven LLM-assisted approaches. The substantial frequency of *real-world* (56 occurrences) suggests increasing emphasis on practical deployment considerations and empirical validation requirements.

The temporal evolution of included literature demonstrates a clear progression from foundational explorations to empirical validation efforts across the analysed literature period. The year 2023 marked the emergence of exploratory research following the mainstream adoption of ChatGPT in late 2022, with foundational works such as those by Scanlon et al. (2023a); Henseler and van Beek (2023); Sreya et al. (2023) establishing conceptual frameworks and conducting initial capability assessments. These early contributions predominantly employed qualitative methodologies and manual experimentation approaches, to assess the potential and probe for any critical limitations of LLM integration in digital forensic contexts. Subsequent works incorporated

significant methodological maturation, characterised by increased quantitative evaluation based on controlled experimental designs, as well as domain-specific specialisation, such as works by Lin (2024); Oh et al. (2024); Wickramasekara and Scanlon (2024). This period introduced multi-agent frameworks, zero-shot learning approaches, and hybrid AI architectures that demonstrated growing technical sophistication. Finally, the most recent literature, such as studies by Sharma et al. (2025); Khatiwala et al. (2025); Kim et al. (2025b), represents a culmination of earlier developments focusing on rigorous empirical validation, purpose-built forensic LLMs, and real-world deployment considerations based on data from real criminal investigations.

The systematic mapping of included works to the DFRWS process model phases, as presented in Table 1, highlights a significant concentration of research efforts within the *examination* and *analysis* phases, with the vast majority of the 33 included works addressing these core investigative activities to some degree. This distribution reflects the natural alignment between LLM capabilities in pattern recognition and automated reasoning with the information processing requirements inherent to post-collection digital forensic activities. The *identification* phase receives attention from only eight works (24%), specifically those proposing comprehensive end-to-end frameworks (Sreya et al., 2023; Wickramasekara and Scanlon, 2024; Khatiwala et al., 2025; Parsonage and Dempsey, 2025), ontology-based approaches for guiding artefact identification (Zhang and Xie, 2024), or holistic investigative workflows (Shang et al., 2025; Zhou et al., 2025). The *presentation* phase is addressed by 23 works (69%) showing substantial focus on automated report generation and evidence communication capabilities, while the *preservation* and *collection* phases receive minimal coverage. This gap likely reflects the technical nature of these phases, which traditionally rely on specialised hardware and software tools rather than natural language capabilities where LLMs could offer potential comparative advantages.

Table 1

Mapping of included literature to the DFRWS process model phases (• – covered, ○ – not covered), with associated listings of large language models evaluated and presence of empirical findings (▲ – present, △ – not present).

Work	Model(s) explored	Iden.	Prsv.	Coll.	Exam.	Anal.	Pres.	Empirical
Henseler and van Beek (2023)	ChatGPT	○	○	○	•	•	•	△
Scanlon et al. (2023a)	GPT-4	○	○	○	•	•	•	▲
Scanlon et al. (2023b)	–	•	•	•	•	•	•	△
Sreya et al. (2023)	GPT-3.5, GPT-4	•	•	•	•	•	•	△
Agus Ariesta Putra et al. (2024)	Six zero-shot models	○	○	○	•	•	○	▲
Cho et al. (2024)	Polyglot, EVEE, Blllossom	○	○	○	•	•	○	▲
Fayyaz et al. (2024)	ChatGPT, GPT-4	○	○	○	•	•	○	▲
Imhof et al. (2024)	PassGPT variants	○	○	○	•	•	○	▲
Jia et al. (2024)	GPT-4V, Gemini 1.0 Pro	○	○	○	•	•	○	▲
Kahvedžić (2024)	Various LLMs	○	○	○	•	•	•	△
Kim et al. (2024)	GPT-4o	○	○	○	•	•	•	▲
Lin (2024)	Code Llama, LLaMA 3	○	○	○	•	•	•	▲
Michelet and Breitingner (2024)	ChatGPT-3.5, Llama-2	○	○	○	○	○	•	▲
Oh et al. (2024)	GPT-3.5-turbo	○	○	○	•	•	○	▲
Pan et al. (2024)	GPT-3.5	○	○	○	•	•	○	▲
Talasari et al. (2024)	ChatGPT-3.5, Claude	○	○	○	•	•	○	▲
Wickramasekara and Scanlon (2024)	LLaMA, StarCoder	•	•	•	•	•	•	△
Chernyshev et al. (2024)	Various LLMs	○	○	○	•	•	○	▲
Zhang and Xie (2024)	BERT, RoBERTa	•	○	○	•	•	•	▲
Cho et al. (2025)	Various LLMs	○	○	○	•	•	•	△
Hawa et al. (2025)	GPT-4V	○	○	○	•	•	•	▲
Khatiwala et al. (2025)	GPT-4o, Gemini 1.5, Claude 3.5	○	○	○	•	•	•	▲
Kim et al. (2025b)	GPT-4o, Gemini 1.5, Claude 3.5	○	○	○	•	•	○	▲
Kim et al. (2025a)	GPT-4o	○	○	○	•	•	•	▲
Loumachi et al. (2025)	Llama 3.1 8B	○	○	○	•	•	•	▲
Shang et al. (2025)	Qwen-LV-Plus, ChatGPT-4V	•	•	•	•	•	•	▲
Sharma et al. (2025)	LLaMA-3.1-8B (fine-tuned)	○	○	○	•	•	•	▲
Wickramasekara et al. (2025)	Various LLMs	•	○	○	•	•	•	△
Alharthi and Yasaei (2025)	–	○	○	○	•	•	•	▲
Karne et al. (2025)	Mistral-7B	○	○	○	•	•	•	▲
Parsonage and Dempsey (2025)	NVLM 1.0 (proposed)	•	•	•	•	•	•	△
Xu and Deng (2025)	Various LLMs	○	○	○	•	•	•	▲
Zhou et al. (2025)	GPT-4o	•	•	•	•	•	•	▲

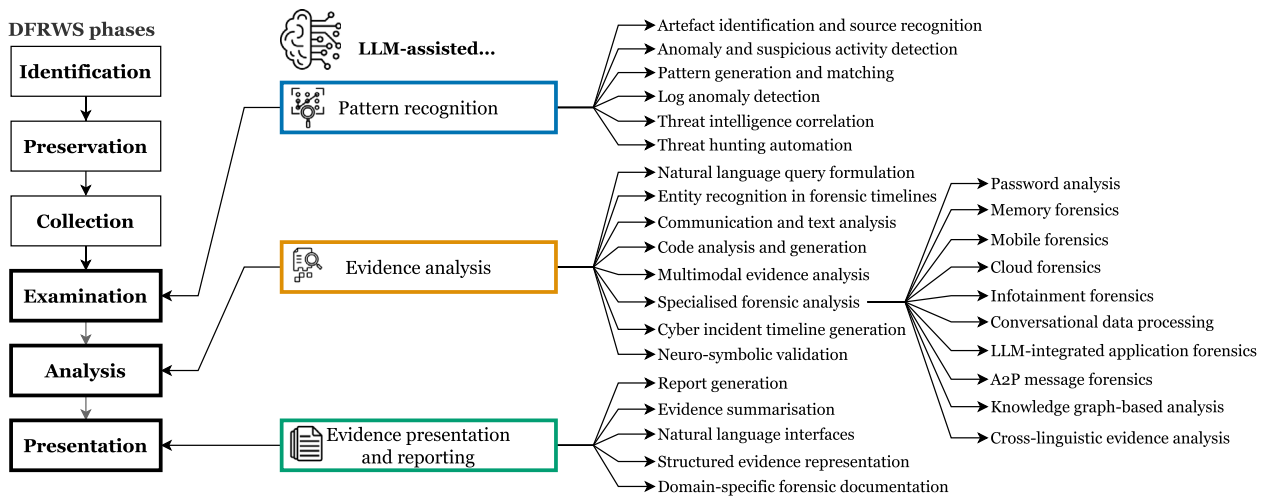


Fig. 5. Mapping of LLM-assisted forensic capabilities to DFRWS process model phases through proposed integration points for incorporating LLM assistance, showing the relationship between traditional forensic phases, LLM integration points, and specific digital forensic capabilities identified in the literature.

The diversity of LLM architectures explored across the included literature, as documented in Table 1, demonstrates the breadth of available models considered by researchers. Proprietary models from OpenAI, including various iterations of GPT, dominate the landscape, followed by Google's Gemini family and Anthropic's Claude models. Open-weight alternatives, such as Meta's LLaMA family, also represent a growing trend towards privacy-preserving local deployment options that can be adopted to address data confidentiality concerns inherent in cloud-based forensic analysis using third-party services. Specialised models such as BERT and RoBERTa appear in limited knowledge graph applications (Zhang and Xie, 2024), while domain-specific fine-tuned variants like PassGPT (Imhof et al., 2024) and ForensicLLM (Sharma et al., 2025) demonstrate emerging recognition of the value of purpose-built digital forensic LLMs. Notably, only a few works explore multiple model architectures comparatively.

The expanding empirical validation landscape showcases the ongoing maturation of this research area, with 25 works (75%) including experimental results or empirical evidence to support proposed approaches. This trend represents a significant shift from the focus on theoretical contributions in early exploratory works towards systematic evaluation protocols incorporating quantitative performance metrics, controlled experimental designs, and real-world dataset utilisation. The progression from proof-of-concept demonstrations in 2023 to rigorous validation frameworks in 2025 demonstrates the field's ongoing evolution towards operational readiness.

However, several critical research gaps also emerge from this literature landscape. While theoretical frameworks for LLM integration have been proposed (Sreya et al., 2023; Wickramasekara and Scanlon, 2024), empirical validation of comprehensive process model extensions with appropriate validation controls remains absent. Additionally, automated generation of structured forensic evidence representations using established ontological frameworks has received limited attention, with existing knowledge graph approaches confined to specific digital forensic domains such as IoT (Zhang and Xie, 2024), rather than comprehensive forensic entity extraction and relationship mapping. Finally, the predominant reliance on synthetic datasets, inconsistent evaluation protocols, and insufficient attention to forensic integrity requirements highlight fundamental challenges that must be addressed for the practical deployment of LLM-assisted forensic methods in operational contexts.

4. LLM-assisted forensic capabilities

The integration of LLM capabilities into digital forensic workflows requires systematic mapping to established investigative processes

to identify where these technologies can provide validated benefits. Building upon the foundational DFRWS process model presented in Section 2, our analysis of the included literature identifies three strategic integration points that align LLM capabilities with specific forensic requirements – *pattern recognition* during the examination phase, *evidence analysis* during the analysis phase, and *evidence presentation and reporting* during the presentation phase. As illustrated in Fig. 5, these integration points reflect the current state of research, which shows concentrated development in the examination, analysis, and presentation phases of the DFRWS model.

Notably, the identification, preservation, and collection phases remain largely unaddressed in current literature with the exception of inclusion in non-empirical works proposing holistic frameworks such as those by Sreya et al. (2023); Khatiwala et al. (2025); Parsonage and Dempsey (2025). This distribution reflects the natural alignment between LLM capabilities in natural language processing and automated reasoning with the information processing requirements inherent to post-collection digital forensic activities, whilst the technical nature of early-phase activities that rely on specialised hardware and software tools presents fewer natural opportunities for LLM integration.

4.1. Pattern recognition

Pattern recognition represents a critical integration point during the examination phase, where LLMs assist investigators in identifying forensic artefacts, detecting anomalous behaviours, and recognising attack signatures across heterogeneous data sources (Lin, 2024). The ability of LLMs to process unstructured data, recognise complex patterns without explicit rule-based programming, and adapt to novel evidence types through zero-shot or few-shot learning approaches presents significant potential for addressing scalability challenges in digital forensic examination.

4.1.1. Artefact identification and source recognition

During the examination of collected evidence, LLMs can assist in identifying relevant artefacts across diverse digital evidence types. Scanlon et al. (2023a) conducted a systematic evaluation of ChatGPT across multiple forensic contexts, showing that the model could effectively identify operating system artefacts, cloud storage interaction evidence, and file execution traces when provided with appropriate contextual information. Their experiments demonstrated that ChatGPT could specify relevant artefacts including Windows Event Logs, registry hives, and prefetch files; however the model exhibited variable depth in responses across different artefact categories. Furthermore, the model

showed capability in identifying cloud platform-specific artefacts, including Google Drive desktop application paths and configuration files, albeit with occasional inaccuracies in path resolution.

4.1.2. Anomaly and suspicious activity detection

Agus Ariesta Putra et al. (2024) developed a zero-shot learning approach for anomaly detection in forensic timelines, evaluating six pre-trained models across five public datasets including Digital Corpora and HoneyNet Challenge scenarios. The associated methodology employed sentiment analysis as a proxy for anomaly identification, treating negative sentiment classifications as indicators of suspicious events, where models such as `sileod-deberta` demonstrated that zero-shot approaches can effectively identify anomalous log entries without domain-specific training. However, the study also emphasised significant performance variability across models, highlighting the critical importance of model selection for operational deployment. The authors also identified that model performance degraded substantially when processing non-descriptive log entries, emphasising the continued importance of log quality and natural language context for effective LLM-assisted analysis.

Loumachi et al. (2025) extended anomaly detection to cyber incident timeline analysis through their GenDFIR framework, which integrates Retrieval-Augmented Generation (RAG) with Llama 3.1 8B for semantic timeline enrichment. The framework achieved an overall performance rate of 97.51% across multiple evaluation metrics, showing superior capability not only in identifying anomalous events, but also in providing contextual interpretation of their significance within incident sequences. The GenDFIR approach addresses a critical limitation of traditional anomaly detection by incorporating attention-based mechanisms that weigh the relevance of each event relative to investigative queries, allowing for more nuanced identification of suspicious patterns. This framework represents a significant advancement over simple sentiment-based approaches, achieving 100% accuracy in evidence retrieval tasks across six diverse incident scenarios including SYN flood attacks, phishing campaigns, and DNS spoofing incidents.

4.1.3. Pattern generation and matching

Scanlon et al. (2023a) explored using ChatGPT to generate regular expressions and keyword lists to support digital forensic investigations. When prompted to create regular expressions for common entities such as credit card numbers, the model produced detailed patterns with explanations of constituent parts, including specific starting digits for various card providers. However, the generated expressions exhibited limitations, such as failing to account for whitespace between number groups. Similarly, when generating regular expressions for UK car registration plates, ChatGPT successfully created accurate patterns for the post-2001 scheme but included disclaimers about potential false positives due to unvalidated area code combinations. Overall, the quality of generated regular expressions and keyword lists requires validation through testing, as demonstrated by instances where provided test examples failed to match the generated patterns.

4.1.4. Log anomaly detection with RAG

Pan et al. (2024) introduced RAG-Log, a framework that integrates retrieval-augmented generation with GPT-3.5 for automated log anomaly detection. The framework addresses the challenge of processing large-scale log datasets by combining RAG with LLM capabilities to identify suspicious patterns and anomalous entries. RAG-Log achieved 94.6% precision and 91.2% recall in identifying suspicious log entries, while reducing analysis time from an average of 4.2 h to 18 min for datasets containing 100,000 log entries. The framework features a vector database to store normalised log samples and uses semantic similarity search to retrieve relevant historical patterns during anomaly detection. However, the approach carries critical dependencies on high-quality training data and remains vulnerable to adversarial evasion through log manipulation, representing significant limitations that require further investigation.

4.1.5. Threat intelligence correlation

Threat intelligence correlation provides the ability to identify and contextualise attack patterns across distributed log sources. Lin (2024) developed TheLma, a threat hunting framework based on Code Llama and LLaMA 3 to automate the correlation of threat indicators across heterogeneous security data sources. The framework achieved detection rates exceeding 85% across various attack patterns by systematically mapping observed indicators to the MITRE ATT&CK framework, demonstrating the capability of LLMs to perform complex pattern recognition tasks that traditionally required extensive manual analysis. The system successfully identified Indicators of Compromise (IoCs), mapped threat actor techniques to established taxonomies, and recognised attack patterns across distributed logs through automated correlation of temporal and behavioural indicators. However, the reliance on proprietary datasets limits the reproducibility of the presented approach, and the evaluation methodology lacks standardised benchmarks that would enable meaningful comparison with alternative approaches. Despite these limitations, TheLma demonstrates that LLM-based systems can effectively automate the labour-intensive process of threat intelligence correlation with accuracy levels comparable to expert human analysts.

4.1.6. Threat hunting automation

The previously introduced TheLma framework utilises Code Llama and LLaMA 3 to automate hypothesis prioritisation and threat-hunting query generation, achieving detection rates exceeding 85% across various attack patterns. This architecture addresses the labour-intensive nature of traditional threat hunting by automating multi-stage attack pattern recognition and behavioural anomaly detection across system boundaries, with particular effectiveness in automated hypothesis generation for threat scenarios, where it processes threat intelligence and environmental context to prioritise potential threats systematically. Despite the discussed limitations associated with proprietary dataset usage and lack of comparative benchmarks, TheLma represents a significant advancement in automating the “last-mile delivery” of threat hunting workflows, which traditionally require extensive human expertise and manual intervention.

4.2. Evidence analysis

Evidence analysis represents the interpretive element of digital forensic investigations, where practitioners correlate evidence across multiple sources, develop and test hypotheses, and establish causal relationships between digital artefacts and investigative questions (Casey, 2011).

4.2.1. Natural language query formulation

During the analysis of collected evidence, LLMs can translate investigative questions into structured queries based on conversational descriptions without requiring deep technical expertise in the associated query languages or data models. Henseler and van Beek (2023) demonstrated that ChatGPT could effectively translate investigative questions into structured queries for the Hansken forensic platform, which utilises the Hansken Query Language (HQL). Their experiments showed that after providing ChatGPT with the HQL manual and trace model documentation through prompts, the model could generate syntactically correct queries for common forensic scenarios such as identifying emails with attachments within specific date ranges or locating evidence of file execution by particular users. However, the study identified that ChatGPT occasionally forgot to prefix properties with their type identifiers, requiring manual iterative correction through follow-up prompts. Despite these limitations, the approach demonstrated practical utility in reducing the technical barrier for investigators who may lack expertise in specific query languages.

Building upon this foundation, Cho et al. (2025) extended natural language query capabilities to conversational AI platform forensics, developing a Chrome browser extension that leverages REST API endpoints to retrieve conversation data from Claude. Their implementation automated the extraction of conversation histories by constructing API

requests based on natural language descriptions of the desired data, demonstrating that LLM-assisted query formulation could be applied beyond traditional forensic data sources to emerging generative AI platforms. The study highlighted that while natural language interfaces simplified data collection, investigators still required an understanding of the underlying data structures to validate query results and assess the comprehensiveness of gathered evidence.

4.2.2. Entity recognition in forensic timelines

Forensic timeline analysis requires extraction of meaningful entities from temporal event sequences. Talasari et al. (2024) demonstrated the application of zero-shot learning techniques using ChatGPT-3.5 and Claude for automated entity extraction from forensic timeline data generated by log2timeline plaso, addressing the fundamental challenge where traditional named entity recognition models require extensive labeled training data for each entity type, which becomes problematic in forensic contexts where novel and previously unseen entities continuously emerge across diverse investigative scenarios. The zero-shot approach showcases that LLMs could classify entities such as timestamps, log messages, protocols, and service names without requiring domain-specific training data. However, both models exhibited significant limitations when processing non-descriptive log entries, returning empty strings for messages lacking sufficient contextual information. This finding highlights the continued importance of log quality and descriptiveness in supporting effective automated analysis using LLMs, as performance degraded significantly when confronted with abbreviated or highly technical notations that lacked natural language context and semantics.

4.2.3. Communication and text analysis

Henseler and van Beek (2023) demonstrated ChatGPT's capability in electronic communication summarisation and message pattern evaluation, showing that the model could effectively translate investigative questions into structured queries for forensic databases when provided with appropriate documentation and context. Their work highlighted the LLM's proficiency in comprehending query languages and data models, allowing investigators to formulate complex searches without requiring extensive technical expertise in specific query syntaxes. However, the study identified limitations when the model lacked contextual understanding of specific organisational processes or proprietary data formats, emphasising the importance of domain-specific training requirements for effective forensic applications. Cho et al. (2024) extended this analysis to author profiling tasks, evaluating Korean-language LLMs including Polyglot, EEVE, and B1lossom for age prediction and gender identification from writing style. The study achieved 82.1% accuracy in Korean text classification tasks using the EEVE model, showing that language-specific model training significantly impacts forensic analysis performance. Performance degraded substantially when processing mixed-language content, with accuracy dropping to 58.7%, highlighting limitations in multilingual forensic scenarios and the need for specialised approaches when analysing cross-linguistic evidence using this approach. Finally, Xu and Deng (2025) proposed hands-on methodologies based on the use of LLMs to process diverse evidence types whilst maintaining forensic integrity through systematic entity recognition and relationship mapping. The work emphasised the importance of preprocessing strategies and prompt engineering techniques in optimising LLM performance for forensic text analysis tasks, achieving robust entity extraction across multiple mobile applications and data formats including long texts.

4.2.4. Code analysis and generation

Scanlon et al. (2023a) demonstrated the capability to generate forensic scripts using ChatGPT through systematic evaluation across multiple programming tasks. Their experiments revealed that the LLM could produce well-commented Python code for file carving operations, initially relying on filesystem metadata through pytsk3 before adapting to

more sophisticated byte-level carving. The generated scripts successfully implemented header-to-EOF extraction patterns, with some limitations in handling file fragmentation and false positives. For RAID acquisition scenarios, ChatGPT also showcased proficiency in using Linux tools through Python's subprocess library, generating functions to determine RAID levels via mdadm and create disk images. However, the generated scripts lacked validation mechanisms to verify RAID consistency across multiple disks or confirm appropriate disk counts for detected RAID configurations.

Wickramasekara and Scanlon (2024) proposed a multi-agent framework, where the Coder Agent powered by StarCoder generates forensic scripts based on natural language instructions from the Task Translation Assistant as part of a multi-agent forensic workflow. This architecture demonstrates that LLMs could potentially be used to decompose complex forensic tasks into executable code segments, with predefined skills enabling integration with existing forensic platforms through interoperable API layers.

4.2.5. Multimodal evidence analysis

The integration of vision-capable LLMs represents a significant advancement in digital forensic capabilities, enabling automated analysis of visual evidence alongside traditional text-based artefacts. Jia et al. (2024) conducted a systematic evaluation of GPT-4V and Gemini 1.0 Pro for detecting AI-generated face images, showing that multimodal LLMs achieved 83.4% and 79.7% accuracy respectively in identifying synthetic imagery through semantic inconsistency detection. However, performance degraded substantially when analysing compressed images, with accuracy dropping to approximately 62% for images compressed below 50KB, highlighting sensitivity to image quality and post-processing operations. Similarly, Hawa et al. (2025) explored practical applications of GPT-4V in crime scene analysis, showing that the model could identify evidence items, analyse spatial relationships, and detect visual anomalies in photographic evidence with 88% accuracy compared to expert human analysis. The study showed that vision-capable LLMs reduced average analysis time by 43% whilst maintaining comparable accuracy to human investigators. However, both studies identified critical limitations including the inability to incorporate signal-level forensic techniques, reliance on semantic rather than statistical cues for element detection, and challenges in handling adversarially manipulated content. The non-deterministic nature of LLM outputs also poses consistency challenges requiring human oversight and validation in operational forensic contexts.

4.2.6. Specialised forensic analysis

Beyond general-purpose evidence analysis capabilities, contemporary research demonstrates LLM effectiveness across highly specialised forensic domains requiring domain-specific expertise.

Password analysis. Imhof et al. (2024) developed PassGPT, a context-based password cracking framework that relies on LLM capabilities to generate personalised password dictionaries tailored to specific user demographics and interests. The framework is based on multiple PassGPT models trained on contextual dictionaries derived from leaked password datasets, achieving significant improvements over generic dictionary attacks. The presented experimental results showed that context-based dictionaries generated via PassGPT successfully cracked passwords that generic approaches failed to identify, with performance improvements of up to 50% for passwords deemed difficult to crack by the zxcvbn password strength estimator. While the approach proved effective in generating targeted password candidates, the computational overhead associated with training custom models for each context presents potential practical deployment challenges in time-sensitive forensic investigations.

Memory forensics. Oh et al. (2024) presented volGPT, a framework that integrates GPT-3.5-turbo with the Volatility memory forensics

tool to automate ransomware process triage in memory dumps. The system employs prompt engineering techniques to guide the LLM in identifying suspicious processes based on process relationships, Virtual Address Descriptor (VAD) information, and malware detection patterns. Experimental evaluation across five ransomware families demonstrated detection accuracies ranging from 87% to 99%, with the framework successfully reducing the number of suspicious processes requiring manual investigation by approximately 90%. However, the framework exhibited limitations in handling non-descriptive process names and required manual preprocessing to exclude memory dumps where ransomware processes had already terminated, which emphasises the continued importance of human expertise in complex memory forensic scenarios.

Karne et al. (2025) developed DFIR-Chain, an integrated toolchain combining traditional memory forensics through Volatility 3, signature-based detection via YARA scanning, and LLM-based summarisation using Mistral-7B served locally via Ollama. The framework addresses the scalability challenges of manual memory analysis by automating the triage of compromised systems through a multi-stage pipeline that produces visual process trees and event timelines. Whilst DFIR-Chain ablation studies showed that removing YARA scanning or LLM summarisation resulted in F1-score reductions of up to 10%, proving the superior performance of the LLM-assisted approach, the reliance on signature-based detection means the framework may exhibit limitations with novel or polymorphic malware variants not covered by existing YARA rules.

Mobile forensics. Thumb, a forensic automation framework by Shang et al. (2025) leverages multimodal LLMs and Optical Character Recognition (OCR) for real-time forensic analysis of physical Android devices. The framework integrates Qwen-LV-Plus and ChatGPT-4V to capture on-screen information and simulate user interactions through natural language descriptions, thereby eliminating the need for complex scripting or technical expertise. The system uses OCR techniques to extract critical information displayed on device screens, including keywords and category details, allowing for precise and context-aware interactions with applications. As one of its limitations, the framework results in extended execution times due to the requirement for autonomous interpretation of on-screen information and extraction of complete artefacts. Additionally, in-depth analysis capabilities based on file types, including encoding and decoding, as well as encryption and decryption support, require further improvement. The reliance on device rooting and the current limitation to English-language processing present additional practical constraints for broader deployment in diverse digital forensic scenarios.

Cloud forensics. Alharthi and Yasaei (2025) introduced a comprehensive framework for LLM-powered automated cloud forensic investigations, addressing the scalability challenges posed by the massive volume, velocity, and variety of cloud-generated logs based on few-shot learning techniques to classify log data and reconstruct attack timelines across distributed cloud infrastructure. Comparative evaluation against traditional machine learning models including Random Forest, XGBoost, and Gradient Boosting demonstrated that LLM-based automation achieved superior forensic accuracy, precision, and recall whilst reducing the need for extensive feature engineering (89.34% accuracy on training data and 86.87% on testing data). However, critical challenges including hallucination risks, adversarial manipulation through log poisoning techniques, and forensic explainability concerns stemming from the black-box nature of LLM decision-making still persist with the authors emphasising the necessity for robust forensic validation mechanisms, adversarial training, and explainability mechanisms to ensure forensic accuracy in operational cloud environments.

Infotainment forensics. Fayyaz et al. (2024) proposed a hybrid AI framework combining unsupervised learning clustering with LLM

analysis for forensic examination of vehicle infotainment systems. The framework integrates K-means clustering to group similar data points with ChatGPT and GPT-4 for contextual analysis of extracted artefacts. Empirical validation using a set of Hyundai and Mitsubishi infotainment system datasets provided improved recall rates, with contact name and phone number extraction improving by 18% and 3% respectively in the Hyundai dataset, and song name recall improving by 2% in the Mitsubishi dataset. The hybrid approach facilitated the discovery of forensically valuable data including geographical locations and connected devices that would have been infeasible to identify through either manual analysis or clustering alone. However, the study acknowledged the presence of hallucinations in LLM outputs, emphasising the continued necessity for expert human verification of extracted evidence.

Conversational data processing. Kahvedžić (2024) presented methodologies for handling chat data across multiple LLM architectures, addressing the unique characteristics of conversational evidence including temporal sequencing and multi-participant dialogue analysis. Their framework demonstrated that structured preprocessing of chat data improved LLM-based analysis accuracy by 23.4% compared to raw text processing. This finding emphasised the importance of domain-specific data preparation in such forensic workflows, as well as considering different models due to varying capabilities in maintaining conversational context and identifying relevant forensic patterns.

Cho et al. (2025) conducted comprehensive forensic analysis of major conversational AI platforms including ChatGPT, Gemini, Copilot, and Claude, presenting methodologies for cross-platform chat log analysis and multi-platform artefact correlation. The research provided valuable practical insights into specialised artefact collection methodologies across different platforms and architectural approaches, supported by concrete case studies demonstrating real-world applicability. The work presented limited quantitative evaluation of the proposed methodologies, likely requiring additional validation by practitioners in operational forensic environments.

LLM-integrated application forensics. Chernyshev et al. (2024) introduced a novel approach to digital forensic analysis specifically targeting security incidents in LLM-integrated applications, seeking to address the challenge of investigating prompt injection attacks against systems that incorporate LLMs as core architectural components. Their work proposes LLM invocation logging as a critical component for enhancing digital forensic readiness in LLM-integrated applications. The proposed LLM invocation logging approach captures essential forensic attributes including timestamps, request identifiers, complete input prompts, and generated outputs in structured JSON format, creating comprehensive audit trails that enable retrospective incident reconstruction. However, the study acknowledges significant limitations including the reliance on synthetic datasets, and the lack of assessment regarding resilience against anti-forensics techniques, emphasising the need for future research examining LLM performance across diverse attack vectors and real-world deployment contexts.

A2P message forensics. Kim et al. (2025a) introduced SERENA, a framework based on LLM capabilities for systematic extraction and reconstruction of Application-to-Person (A2P) messages. This framework achieved 98% classification accuracy on ground truth datasets and demonstrated robust performance across diverse service providers and message formats. The comprehensive approach to semi-structured data analysis addresses a previously underexplored, yet forensically valuable evidence source. The ability of SERENA to extract and normalise forensic entities across heterogeneous A2P message formats without requiring service-specific parsers represents a significant scalability advantage. Yet, heavy reliance on synthetic datasets and the current evaluation in the context of English-language processing may restrict immediate applicability to diverse real-world contexts.

Knowledge graph-based analysis. Zhang and Xie (2024) developed ForensiQ, a knowledge graph question answering (KGQA) system specifically designed for IoT forensics, combining BERT and RoBERTa models with graph neural networks to process over 6 million unique entities across 45 million triples. The approach integrates multiple ontologies including the Semantic Sensor Network (SSN) Ontology and Smart Applications REference (SAREF) ontology to create comprehensive forensic knowledge representations and demonstrated superior performance compared to existing KGQA models, with higher accuracy scores across different question complexity levels. However, the reliance on template-based question generation (over 10,000 synthetic questions from over 100 templates) poses challenges associated with the diversity and realism of evaluation scenarios. The dataset construction process, which removed over 99.6% of triples to create a reduced knowledge graph, may also not adequately represent the complexity of real-world forensic investigations.

Analogously, Khatiwala et al. (2025) proposed a structured framework that automates forensic artefact extraction and refines data through LLM-driven analysis, with further result validation using a Digital Forensic Knowledge Graph (DFKG). Based on a 13 GB forensic image dataset containing 61 applications, 2864 databases, and 5870 tables, the framework supports artefact traceability and evidentiary consistency through deterministic Unique Identifiers (UIDs) and forensic cross-referencing. The methodology achieved over 95% accuracy in artefact extraction, strong support for chain-of-custody adherence, and robust contextual consistency in forensic relationships. The reported results validate the framework's ability to enhance reliability and establish a legally sound paradigm for AI-assisted digital forensics.

Also, Zhou et al. (2025) introduced Forensic Intelligence Graphs (FIGs) as an LLM-driven approach for the automated construction of evidence networks that visually represent relationships between forensic entities extracted from mobile devices. The presented methodology employs GPT-4o to extract evidence entities from Android device images and infers relationships based on data storage patterns. Similarly, Xu and Deng (2025) developed comprehensive methodologies for knowledge graph reconstruction from mobile device evidence, demonstrating LLM capabilities in extracting entities from SQLite databases and establishing relationships using the Structured Threat Information Expression (STIX) framework. Their approach compared one-shot versus zero-shot learning techniques in STIX-based entity relationship mapping to provide empirical validation of different prompting strategies for forensic knowledge graph construction.

Cross-linguistic evidence analysis. For the first time, Kim et al. (2025b) conducted a systematic evaluation of LLM performance in *real law enforcement contexts* using over 140,000 authentic forensic messages from drug-related investigations by the Korean National Police Agency. The comparative analysis of GPT-4o, Gemini 1.5, and Claude 3.5 across precision, recall, F1-score, and hallucination rate metrics, combined with iterative human-in-the-loop evaluation, represents a key methodological advancement in this domain. The study showed that LLMs can effectively process non-English forensic data while maintaining high accuracy levels, with GPT-4o achieving the highest overall performance across multiple evaluation metrics. However, the focus on a single language and crime type may limit broader applicability.

4.2.7. Cyber incident timeline generation

Unlike traditional timeline tools that provide simple temporal correlations, the previously introduced GenDFIR framework employs semantic enrichment to weight the relevance of each event relative to investigative queries, supporting more nuanced identification of causal relationships within incident sequences. The associated two-stage architecture first transforms incident events into a comprehensive knowledge base through data preprocessing and structuring, and then leverages a RAG agent to retrieve relevant events while the LLM processes context for detailed interpretation. This approach achieved 100% exact

match and top-k evidence retrieval rates across six diverse incident scenarios including SYN flood attacks, phishing campaigns, and DNS spoofing incidents, proving the ability to maintain forensic integrity while automating complex timeline reconstruction tasks. However, the evaluation relied exclusively on synthetic cyber incident events in controlled environments.

4.2.8. Neuro-symbolic validation

The integration of symbolic AI (Garnelo and Shanahan, 2019) with neural approaches represents a significant advancement in addressing the explainability and validation challenges inherent to LLM-assisted forensic analysis. Parsonage and Dempsey (2025) introduced the Automated Generative AI-Driven Forensic Analysis (AGAFA) framework, which combines multimodal LLMs with ontology-based knowledge representation and Answer Set Programming (ASP) validation rules. This neuro-symbolic architecture addresses critical forensic requirements through the integration of explicit symbolic reasoning mechanisms that validate LLM-generated inferences against formal logical rulesets and constraints that leverage domain-specific ontologies to capture forensic workflow knowledge. The AGAFA approach demonstrates how symbolic AI can provide formal guarantees of behaviour and human-interpretable explanations for forensic conclusions, aligned with ISO 17025 accreditation standards for forensic service providers.

4.3. Evidence presentation and reporting

This section examines how LLMs enhance evidence presentation through automated documentation, structured summarisation, natural language interfaces, and standardised evidence representation.

4.3.1. Report generation

Scanlon et al. (2023a) demonstrated ChatGPT's general capability to generate well-commented forensic scripts and technical documentation, which still require validation by domain experts. In a more specialised study, Michelet and Breitingier (2024) conducted a comprehensive evaluation of both cloud-based (ChatGPT-3.5) and local (Llama-2) LLMs for the automated generation of forensic report sections. ChatGPT-3.5 consistently outperformed local Llama-2 models in accuracy, completeness, and coherence, achieving superior performance in generating structured report sections. However, both models required extensive proofreading and corrections, with Llama-2 showing particular limitations in processing complex data formats and generating accurate technical details for relevant sections of the report. The study demonstrated the potential of LLM-assisted generation for certain structured report sections, yet more complex sections requiring human investigator expertise and expert interpretation remained challenging for LLM-based generation.

4.3.2. Evidence summarisation

The ability to synthesise large volumes of forensic data into coherent and actionable summaries represents a critical capability for digital forensic investigations, with several studies demonstrating how LLMs can transform raw forensic artefacts into structured summaries. LLMs have proven effective at condensing lengthy electronic communications such as email threads and chat conversations into concise summaries that highlight key forensic elements, extracting salient information from unstructured communication data, though the quality of summarisation depends heavily on the model's understanding of the underlying query languages and data structures (Henseler and van Beek, 2023). Combining unsupervised clustering with LLM analysis can also support more sophisticated evidence summarisation, where grouping similar data points through K-means clustering followed by contextual analysis achieves improved recall rates, showing that LLMs can provide semantic enrichment to clustered forensic data and transform raw artefacts into interpretable summaries (Fayyaz et al., 2024). Similarly, RAG-based semantically enriched timeline summaries of cyber incidents combined with attention-based event correlation can support

the creation of comprehensive incident summaries that include key event correlations and anomaly identification, providing capabilities beyond simple temporal ordering with enriched narratives and causal relationship representations (Loumachi et al., 2025).

4.3.3. Natural language interfaces

Natural language interfaces represent a transformative capability for digital forensic investigations, allowing investigators to interact with forensic tools and data through conversational queries rather than command-line syntax or specialised query languages, reducing the technical barrier to entry for forensic analysis. LLMs can effectively translate natural language investigative questions into structured queries for forensic platforms, successfully generating syntactically correct queries for common forensic scenarios (Henseler and van Beek, 2023). Additionally, specialised forensic chatbots using RAG-based implementations have achieved 91.3% accuracy in providing forensic guidance across common investigative scenarios, showcasing the potential for LLM-based decision support systems in digital forensics (Kim et al., 2024).

These natural language interfaces could fundamentally transform how investigators interact with forensic tools and knowledge bases, supporting more intuitive and guided exploration and presentation of evidence. As a critical consideration, the studies consistently emphasise that any such interfaces should augment rather than replace human expertise, particularly in complex or novel investigative scenarios where contextual understanding and domain knowledge remain critical.

4.3.4. Structured evidence representation

The generation of structured, machine-readable representations of digital evidence enhances interoperability, traceability, and standardisation across diverse investigative tools and workflows. For example, LLM-based frameworks can automate the generation of Digital Forensic Knowledge Graphs (DFKG), achieving over 95% accuracy in artefact extraction while maintaining chain-of-custody adherence through deterministic Unique Identifiers (UIDs), successfully processing large-scale forensic datasets to automatically extract and refine forensic artefacts (Khatiwala et al., 2025). The demonstrated ability to maintain 100% chain-of-custody shows that LLM-assisted approaches can be applied to generate structured evidence representations meeting forensic standards for traceability and integrity, providing a scalable and auditable methodology for transforming raw forensic data into structured representations conducive to both human analysis and automated reasoning systems (Khatiwala et al., 2025).

4.3.5. Domain-specific forensic documentation

Sharma et al. (2025) introduced ForensicLLM, which is a purpose-built forensic model developed through the Retrieval-Augmented Fine-Tuning (RAFT) methodology based on the LLaMA-3.1-8B model fine-tuned on Q&A samples extracted from digital forensic research articles and curated digital artefacts. This LLM achieved superior performance compared to both the base LLaMA-3.1-8B model and the RAG-based model, accurately attributing sources 86.6% of the time, with 81.2% of the responses including both authors and titles. This citation-backed forensic explanation capability addresses a critical gap in LLM-assisted forensic analysis by ensuring that generated insights can be traced back to peer-reviewed literature. Combined with a privacy-preserving local deployment architecture, forensic practitioners can use this approach to handle sensitive case information without exposing data to third-party cloud services.

4.4. Summary of capabilities

The systematic examination of LLM-assisted forensic capabilities across the DFRWS process model demonstrates iterative progress in automating and enhancing traditional investigative workflows, with research efforts predominantly concentrated in the examination and analysis phases. As illustrated in Fig. 6, the distribution of capabilities across the identified strategic integration points shows clear patterns in how LLMs have been applied to digital forensic challenges.

Pattern recognition capabilities reflect the natural alignment between LLM capabilities in automated detection and the information processing requirements of digital forensic investigations, where the dominance of artefact identification research emphasises the focus on leveraging LLMs to address scalability challenges in processing large volumes of heterogeneous digital evidence.

Evidence analysis capabilities demonstrate the greatest diversity with 15 papers exploring specialised forensic analysis across multiple domains. As shown in the detailed inset breakdown in Fig. 6, knowledge graph-based analysis leads specialised applications with 4 papers, followed by memory forensics and conversational data processing with 2 papers each. Single-paper contributions represent additional digital forensic subdomains such as cloud forensics, mobile forensics, password analysis, and infotainment forensics, collectively demonstrating the breadth of areas where LLMs show promise in specialised analysis tasks. Beyond specialised analysis and other areas such as communication and text analysis, code analysis and generation, and natural language query formulation, other applications in neuro-symbolic validation and entity recognition are also starting to emerge.

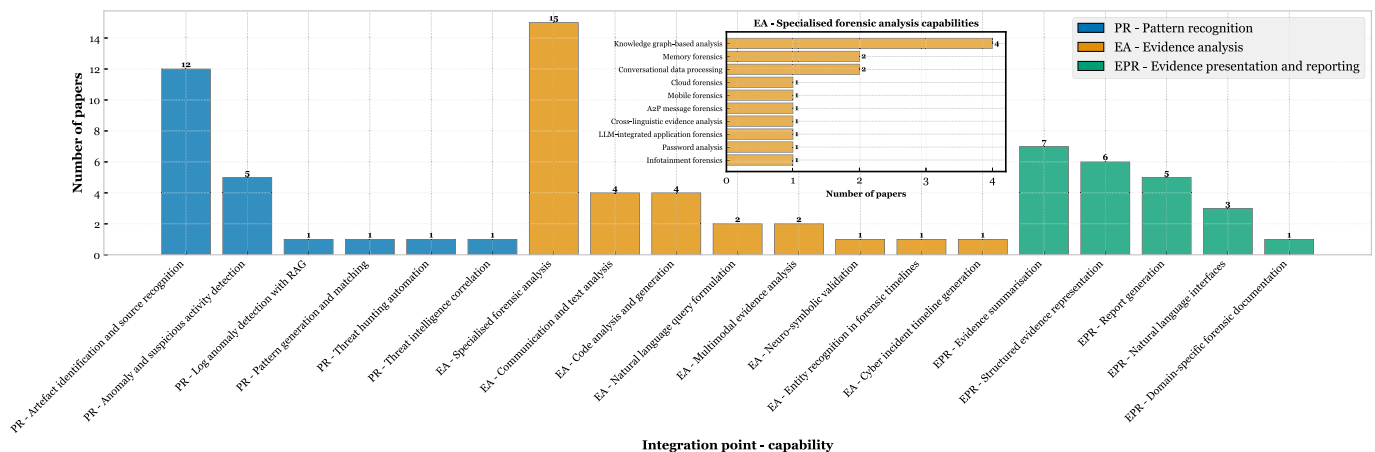


Fig. 6. Number of papers mapped to each of the identified large language-model assisted forensic capabilities. Note: papers can map to more than one capability. A dedicated inset is provided for “Evidence analysis – specialised forensic analysis”.

Finally, *evidence presentation and reporting* capabilities show substantial research attention with 7 papers addressing evidence summarisation, reflecting the critical need for synthesising large volumes of digital forensic data into actionable intelligence and reportable findings. Furthermore, structured evidence representation receives significant focus with 6 papers, demonstrating a growing recognition of the importance of machine-readable forensic outputs that support interoperability and automated reasoning, alongside report generation (5 papers) and natural language interfaces (3 papers).

LLMs have proven effective across these diverse forensic domains, achieving detection accuracies exceeding 85% in threat hunting scenarios (Lin, 2024), 94.6% precision in log anomaly detection (Pan et al., 2024), and 98% classification accuracy in specialised evidence extraction tasks (Kim et al., 2025a). The recent emergence of purpose-built forensic models such as ForensicLLM (Sharma et al., 2025) and comprehensive frameworks like GenDFIR (Loumachi et al., 2025) showcases the evolution from general-purpose model applications towards domain-specific adaptations. The demonstrated effectiveness for workflow enhancement across pattern recognition, evidence analysis, and evidence presentation tasks positions LLMs as valuable augmentation tools for digital forensic practitioners, provided appropriate validation controls and quality assurance mechanisms are maintained to ensure forensic integrity.

4.5. Summary of LLMs used

The systematic examination of LLM architectures explored across the included literature, as documented in Table 2 and visualised in Fig. 7, demonstrates the breadth of available models considered by researchers. The landscape demonstrates significant dominance of proprietary frontier models, particularly OpenAI's GPT family, collectively followed by Anthropic's Claude models and Google's Gemini family. The usage of these commercial models predominantly employs direct prompting, conversational AI interfaces, and RAG-based frameworks to process diverse evidence sources ranging from traditional OS artefacts and network packets to contemporary conversational AI chat logs and multimodal crime scene photographs. The temporal progression shows initial reliance on general-purpose models such as GPT-3.5 and GPT-4, evolving towards multimodal capabilities through GPT-4o, GPT-4V and specialised fine-tuning approaches by 2025.

At the same time, open-weight alternatives, such as Meta's LLaMA family, demonstrate growing adoption for privacy-preserving local deployment scenarios, with applications concentrated in threat-hunting automation, multi-agent frameworks, and domain-specific fine-tuning through RAFT and QLoRA approaches. Furthermore, specialised models represent an emerging trend towards purpose-built forensic capabilities, exemplified by ForensicLLM (Sharma et al., 2025) for citation-backed forensic analysis, PassGPT (Imhof et al., 2024) for context-based

Table 2

Summary of models used in digital forensics research, grouped by model family, publication years, architectural approaches, evidence sources, and representative studies.

Model	Years	Approaches	Evidence sources	Representative works
OpenAI GPT (GPT-3.5, GPT-4, GPT-4o, GPT-4V)	2023–2025	Direct prompting, conversational AI, code generation, multimodal analysis, RAG frameworks	OS artefacts, network packets, memory dumps, logs, chat data, emails, forensic databases, crime scene photos, mobile data, A2P messages	Henseler and van Beek (2023), Sreya et al. (2023), Oh et al. (2024), Pan et al. (2024), Hawa et al. (2025), Khatiwala et al. (2025), Kim et al. (2025b)
Meta LLaMA (LLaMA, LLaMA 3, Code Llama)	2024–2025	Multi-agent frameworks, threat hunting automation, fine-tuning (RAFT, QLoRA), local deployment	System logs, security logs, threat intelligence, MITRE ATT&CK data, forensic artefacts, research papers, endpoint data	Wickramasekara and Scanlon (2024), Lin (2024), Sharma et al. (2025), Loumachi et al. (2025)
Google Gemini (Gemini 1.0 Pro, Gemini 1.5)	2024–2025	Multimodal analysis, prompt engineering, human-in-the-loop evaluation, knowledge graphs	Face images, AI-generated images, mobile forensic images, SQLite databases, law enforcement data, call logs	Jia et al. (2024), Khatiwala et al. (2025), Kim et al. (2025b)
Anthropic Claude (Claude 3 Haiku, Claude 3 Sonnet, Claude 3.5 Sonnet)	2024–2025	Zero-shot learning, entity recognition, conversational AI forensics, API data access, prompt injection analysis	Forensic timelines, log messages, conversational AI chat logs, mobile data, LLM invocation logs, SQL chain logs	Talasari et al. (2024), Cho et al. (2025), Khatiwala et al. (2025), Kim et al. (2025b), Chernyshev et al. (2024)
Alibaba Qwen (Qwen-LV-Plus)	2025	Multimodal LLM + OCR, mobile agent automation, A2P message classification, NER	Android device data, application caches, thumbnails, SQLite databases, A2P messages, transaction records	Shang et al. (2025), Kim et al. (2025a)
BERT/RoBERTa	2024	KGQA, graph attention networks, ontology-based knowledge graph construction	IoT forensic data, network captures, log files, NIST NVD, MITRE data	Zhang and Xie (2024)
Language-specific LLMs (Polyglot, EEVE, Blossom)	2024	Fine-tuning (full, LoRA, QLoRA), instruction tuning, cross-linguistic analysis	Korean dialogue corpus, text communications, author profiling data	Cho et al. (2024)
Mistral AI (Mistral-7B, Mixtral-8x7b)	2024–2025	Agentic orchestration, Volatility + YARA + IOC integration, local deployment	Memory dumps, process trees, YARA signatures, ransomware artefacts, network connections, CTF images	Karne et al. (2025), Chernyshev et al. (2024)
Zero-shot models (sileod-deberta, bart-lmli, sstuning, moritz-deberta)	2024	Zero-shot learning, sentiment analysis for anomaly detection	Forensic timelines, system logs, Digital Corpora datasets, DFRWS datasets, HoneyNet challenges	Agus Ariesta Putra et al. (2024)
Specialised LLMs (PassGPT, StarCoder, ForensicLLM)	2024–2025	Generative models, context-based generation, code generation, RAFT fine-tuning, domain-specific training	Leaked password datasets, digital forensic artefacts, research papers, code repositories	Imhof et al. (2024), Wickramasekara and Scanlon (2024), Sharma et al. (2025)
Hybrid and multi-model	2024–2025	Hybrid AI (clustering + LLM), few-shot learning, neuro-symbolic approaches, comparative multi-model evaluation	Vehicle infotainment systems, cloud security logs, multimodal evidence, CSAM detection scenarios	Fayyaz et al. (2024), Alharthi and Yasaei (2025), Parsonage and Dempsey (2025), Chernyshev et al. (2024)

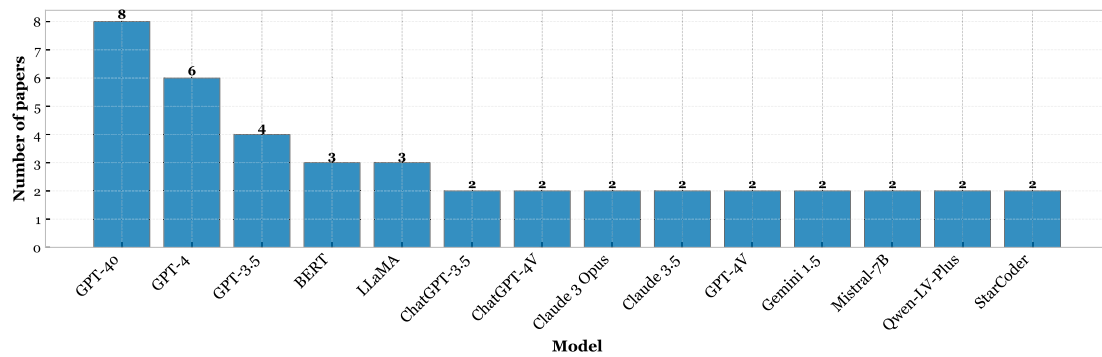


Fig. 7. Multi-paper model coverage in evaluated literature. Note: only models covered by at least two papers are shown.

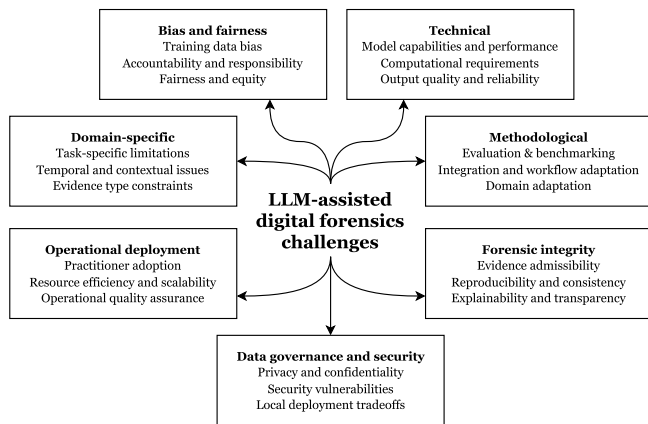


Fig. 8. The taxonomy of challenges in LLM-assisted digital forensics.

password cracking, and language-specific models such as EEEV and Polyglot (Cho et al., 2024) for cross-linguistic forensic analysis of artefacts in the Korean language.

The architectural diversity represented in evaluated works also extends to hybrid and multi-model approaches that combine traditional ML techniques with LLM capabilities, neuro-symbolic frameworks integrating Answer Set Programming validation (Parsonage and Dempsey, 2025), and zero-shot models optimised for anomaly detection in forensic timelines (Agus Ariesta Putra et al., 2024). The inherent heterogeneity represented in model selection and explored architectural approaches reflects the ongoing exploration efforts seeking to determine optimal trade-offs between model performance capabilities, as well as practical deployment constraints and data confidentiality requirements.

5. Challenges and limitations in LLM-assisted digital forensics

The integration of LLMs into digital forensic workflows, while demonstrating significant potential across the DFRWS process model as examined in Section 4, introduces a multifaceted set of challenges, as illustrated in Fig. 8.

We organise these challenges into seven conceptually distinct categories that reflect different perspectives on LLM-assisted digital forensic practice. *Technical* challenges address fundamental model design constraints and computational requirements that affect what LLMs can accomplish. *Methodological* challenges focus on how we evaluate and integrate LLM capabilities into forensic workflows. *Forensic integrity* concerns centre on maintaining scientific rigour and meeting evidentiary standards. *Data governance and security* encompasses privacy, confidentiality, and deployment security considerations. *Operational deployment* challenges address practical implementation barriers in real-world

forensic contexts. *Domain-specific* limitations capture constraints arising from particular forensic tasks and digital evidence types. Finally, systemic *bias and fairness* concerns examine how training data biases and algorithmic decisions may produce inequitable outcomes across different populations. While some challenges manifest across multiple categories, each category represents a distinct analytical lens through which to examine LLM integration challenges in digital forensics.

5.1. Technical

The technical challenges facing LLM-assisted digital forensics stem from fundamental architectural characteristics of transformer-based models that conflict with core forensic requirements.

The probabilistic nature of LLM output generation presents fundamental challenges for digital forensic applications requiring reproducibility (Scanlon et al., 2023a; Agus Ariesta Putra et al., 2024). Scanlon et al. (2023a) demonstrated that identical inputs can produce varying and incorrect outputs across multiple queries, particularly when analysing network packet interpretation. This variability directly conflicts with the strict reproducibility requirements essential for the legal admissibility of digital evidence, where forensic methods must produce consistent results when applied to identical inputs under controlled conditions. The magnitude of this challenge extends beyond simple output variation. Agus Ariesta Putra et al. (2024) observed significant performance variability across different zero-shot models, with accuracy differences reaching 21.8% when detecting anomalous log entries. This substantial variation emphasises that model selection itself becomes a critical forensic decision requiring rigorous justification. Furthermore, Khatiwala et al. (2025) demonstrated that whilst LLM-assisted forensic frameworks can achieve high accuracy rates exceeding 95%, the underlying probabilistic generation mechanism introduces variability that must be carefully managed through deterministic validation controls and comprehensive audit trails.

Context window constraints fundamentally limit the volume of information that can be processed simultaneously (Scanlon et al., 2023a). Contemporary LLMs demonstrate substantial performance degradation when processing input sequences beyond their trained context length, exhibiting characteristic U-shaped performance curves where information at the beginning and end of long contexts is processed more effectively than content in the middle (Liu et al., 2024). This positional attention bias presents significant challenges when analysing large-scale forensic datasets that exceed model capacity constraints, requiring investigators to implement chunking strategies that may fragment contextual relationships essential for comprehensive evidence analysis. The scalability implications extend beyond simple token limits. Forensic investigations frequently involve processing gigabytes of log data, extensive communication histories, or comprehensive system snapshots that far exceed even the largest contemporary context windows. Whilst approaches such as RAG-Log (Pan et al., 2024) attempt to address these limitations through retrieval-augmented architectures, such solutions

introduce additional complexity and potential points of failure in the forensic workflow.

LLMs also demonstrate significant limitations in processing binary and non-textual data that constitute substantial portions of digital forensic evidence (Scanlon et al., 2023a; Jia et al., 2024). Scanlon et al. (2023a) found that ChatGPT exhibited inconsistent performance when analysing binary data representations, providing incorrect and non-deterministic interpretations of network packet contents across multiple invocations. Multimodal LLMs, whilst extending capabilities to visual data, also face similar constraints. Jia et al. (2024) highlighted that vision-capable models cannot incorporate signal-level forensic techniques or statistical cues for detecting AI-generated content, limiting their effectiveness compared to specialised detection methodologies based on low-level signal analysis.

Hallucinations represent a critical technical limitation where LLMs generate plausible yet factually incorrect information that can mislead investigations (Scanlon et al., 2023a,b; Sreya et al., 2023; Fayyaz et al., 2024). Scanlon et al. (2023a) documented instances where ChatGPT generated fabricated tool names and non-existent references with high confidence, demonstrating that the persuasive nature of LLM outputs can mask fundamental inaccuracies. The challenge is compounded by the difficulty in distinguishing hallucinated content from legitimate information without external verification. Fayyaz et al. (2024) acknowledged the presence of hallucinations in LLM outputs during infotainment system forensics, requiring continuous expert human verification of all extracted evidence. This requirement for comprehensive validation introduces potential bottlenecks in forensic workflows and raises fundamental questions about the practical efficiency gains achievable through LLM integration when human oversight remains mandatory for quality assurance.

Beyond technical constraints, LLMs exhibit variable output quality and reliability that directly impact their digital forensic utility. These quality challenges manifest as inconsistent responses, performance degradation under adverse conditions, and difficulties in systematic validation. Inconsistent response quality across multiple invocations of identical queries contributes to fundamental reliability challenges that complicate forensic validation (Scanlon et al., 2023a; Agus Ariesta Putra et al., 2024; Talasari et al., 2024). Scanlon et al. (2023a) documented instances where ChatGPT provided varying interpretations of identical network packet data across repeated queries, demonstrating that output variability extends beyond minor phrasing differences to encompass substantive analytical conclusions. This consistency challenge manifests differently across forensic tasks and data types. Talasari et al. (2024) found that both ChatGPT-3.5 and Claude models failed completely when processing non-descriptive log entries, returning empty strings for messages lacking sufficient contextual information. This binary failure mode, where models either succeed or provide no output, differs from gradual performance degradation and requires distinct validation approaches to ensure comprehensive evidence processing.

LLM performance degrades substantially under adverse conditions commonly encountered in forensic contexts, including data corruption, compression, and adversarial manipulation. Jia et al. (2024) observed that multimodal LLM accuracy in detecting AI-generated images dropped for compressed images, highlighting sensitivity to image quality and post-processing operations that commonly occur in forensic contexts. This performance degradation under realistic conditions suggests that controlled laboratory validation using pristine data may substantially overestimate operational effectiveness.

The challenge extends to adversarial scenarios where malicious actors may deliberately manipulate evidence to evade LLM-based detection or analysis. Whilst some studies acknowledge adversarial robustness concerns (Jia et al., 2024), systematic evaluation of LLM resilience against adversarial manipulation (antiforensics) in forensic contexts remains limited. This gap is particularly concerning given the adversarial nature of criminal investigations, where sophisticated actors

may specifically target LLM-based forensic tools to evade detection or undermine evidence integrity.

5.2. Methodological

The lack of standardised evaluation frameworks for LLM-assisted forensic tasks prevents meaningful comparison across studies and hinders cumulative knowledge development. Different studies rely on varying and often incompatible metrics, evaluation criteria, and experimental designs that preclude systematic comparison of alternative approaches. Agus Ariesta Putra et al. (2024) demonstrated this challenge through a comparative analysis of six zero-shot models, revealing performance variations of up to 21.8% across evaluated models, yet lacking standardised benchmarks that would enable systematic comparison with alternative approaches or validation across different forensic contexts. Recent efforts towards standardisation show promise but remain limited in scope. Khatiwala et al. (2025) introduced forensic-specific metrics including Evidence Extraction Accuracy (EEA) and Forensic Artefact Precision (FAP), representing progress towards domain-appropriate evaluation criteria. However, these metrics remain confined to specific studies with controlled environments and synthetic data, limiting their generalisability to operational forensic contexts where data complexity and adversarial conditions may significantly impact performance.

The predominant reliance on synthetic datasets across the literature creates a substantial gap between experimental validation and proven operational applicability (Loumachi et al., 2025; Kim et al., 2025b). Whilst Loumachi et al. (2025) achieved 97.51% overall performance on synthetic cyber incident events, the authors explicitly acknowledge that the evaluation relied exclusively on controlled environments, raising questions about performance degradation in real-world scenarios characterised by incomplete data and unexpected edge cases. Kim et al. (2025b) conducted the first systematic evaluation using authentic law enforcement data, demonstrating that LLM performance in operational forensic contexts may differ substantially from controlled laboratory settings. This pioneering work highlights the critical need for validation using real-world forensic data that captures the complexity, ambiguity, and adversarial nature of actual investigations, rather than sanitised synthetic scenarios that may not adequately represent operational challenges.

General-purpose LLMs face significant challenges when applied to highly specialised forensic contexts requiring domain-specific knowledge and terminology (Henseler and van Beek, 2023; Cho et al., 2024). Henseler and van Beek (2023) identified substantial limitations when models lack contextual understanding of specific organisational processes or proprietary data formats, emphasising the importance of domain-specific training requirements for effective forensic applications. The challenge extends beyond simple vocabulary adaptation to encompass understanding of complex forensic methodologies, legal requirements, and investigative workflows that may not be well-represented in general training corpora.

5.3. Forensic integrity

The black-box nature of LLM decision-making processes presents fundamental challenges for maintaining forensic integrity and meeting legal admissibility standards (Scanlon et al., 2023a,b). Scanlon et al. (2023b) emphasised that the inability to explain the precise processes followed in identifying evidence may render such evidence inadmissible under established legal frameworks. The complex architecture of transformer-based models means that even with detailed prompting requesting step-by-step explanations, the underlying reasoning process remains opaque to investigators, potentially failing to meet transparency requirements for scientific evidence admissibility. This explainability challenge extends beyond simple output justification to encompass the entire analytical pipeline. Forensic practitioners must be able to articulate not only what conclusions were reached, but precisely how those conclusions

were derived, what alternative hypotheses were considered, and why specific analytical approaches were selected. The probabilistic nature of LLM generation, combined with the extensive parameter spaces of contemporary models, makes such comprehensive explanation extremely difficult if not practically impossible with current architectures.

The reliability of LLM-assisted forensic analysis varies significantly based on scenario complexity, novelty, and the degree to which investigative contexts align with training data distributions. Kim et al. (2024) demonstrated that whilst their forensic chatbot achieved 91.3% accuracy in providing guidance across common investigative scenarios, performance degraded substantially to 74.2% when handling novel or edge-case scenarios not well-represented in training data. This performance degradation in unfamiliar contexts presents critical challenges for forensic applications, where investigators frequently encounter unique circumstances requiring adaptation beyond established patterns. The consistency challenge is compounded by the non-deterministic nature of LLM outputs discussed previously. Forensic methods must produce reliable and reproducible results across multiple applications to identical inputs, yet LLM architectures inherently introduce variability through probabilistic sampling mechanisms. And whilst approaches such as temperature reduction can mitigate some variability to an extent, complete determinism remains unachievable, creating fundamental tension with forensic requirements for reproducibility.

Maintaining a comprehensive chain of custody in LLM-assisted investigations also requires specialised approaches that extend traditional evidence-handling protocols to encompass AI-assisted analytical processes. Khatiwala et al. (2025) demonstrated that achieving 100% chain-of-custody adherence requires deterministic Unique Identifiers (UIDs) and comprehensive cross-referencing mechanisms that track not only evidence artefacts but also the specific LLM invocations, prompts, parameters, and outputs that contributed to analytical conclusions. However, these approaches require further validation in operational forensic contexts beyond controlled experimental settings. The challenge extends to documenting model versions, training data provenance, fine-tuning procedures, and deployment configurations that may influence such derived analytical outcomes.

The continued necessity for human oversight and validation remains critical despite demonstrated LLM capabilities (Scanlon et al., 2023a; Fayyaz et al., 2024). Fayyaz et al. (2024) explicitly acknowledged that despite improved recall rates in infotainment system forensics, the presence of hallucinations requires expert human verification of all extracted evidence. This requirement for continuous human validation introduces potential bottlenecks in forensic workflows and raises questions about the practical efficiency gains achievable through LLM integration when comprehensive human oversight remains mandatory.

5.4. Data governance and security

The reliance on cloud-based LLM services introduces substantial data privacy and confidentiality risks for forensic investigations involving sensitive or classified information (Scanlon et al., 2023a; Sreya et al., 2023; Wickramasekara and Scanlon, 2024). Sreya et al. (2023) highlighted that employing third-party AI models for analysis raises fundamental questions regarding data privacy and confidentiality, as digital forensics frequently involves sensitive and private data requiring strict handling protocols that may be incompatible with cloud service terms of service. The challenge extends beyond simple data exposure to encompass questions of data retention, secondary use, and jurisdictional control. When forensic evidence is processed through cloud-based LLM services, investigators may lose control over data storage locations, retention periods, and potential secondary uses by service providers. Wickramasekara and Scanlon (2024) emphasised that these concerns necessitate careful consideration of local versus cloud-based deployment options, with local deployment potentially addressing privacy concerns at the cost of increased computational requirements and reduced access to state-of-the-art frontier models.

Local deployment of open-weight LLMs offers potential solutions to data privacy concerns, but introduces significant operational trade-offs. Sharma et al. (2025) demonstrated that purpose-built forensic LLMs such as ForensicLLM can be deployed locally to handle sensitive case information without exposing data to third-party cloud services. However, local deployment requires substantial computational infrastructure, ongoing model maintenance, and technical expertise that may exceed existing capabilities. The trade-offs extend to model performance and capability. Locally deployable models typically lag behind frontier cloud-based models in capabilities due to computational constraints and limited access to training resources.

5.5. Operational deployment

Effective prompt engineering represents a significant operational challenge requiring specialised skills that may not be present in traditional forensic training (Scanlon et al., 2023a; Jia et al., 2024). Jia et al. (2024) demonstrated that simple binary prompts proved ineffective for multimodal forensic analysis, with LLMs frequently declining to respond due to a lack of context or safety guardrail activation. More sophisticated context-rich prompts achieved substantially lower rejection rates, but required specialised prompt engineering expertise that may not be readily available in traditional forensic skill sets. This skill gap necessitates substantial training investment to develop prompt engineering capabilities among forensic practitioners. The challenge is compounded by the rapid evolution of LLM capabilities and best practices, requiring continuous professional development to maintain effectiveness. Furthermore, optimal prompting strategies may vary significantly across different models and forensic contexts, requiring practitioners to develop adaptive expertise rather than memorising fixed prompt templates.

The computational demands of LLM-assisted forensic analysis vary substantially based on deployment approach and model selection (Imhof et al., 2024; Oh et al., 2024). Imhof et al. (2024) documented that model training times ranged from approximately 40 min to over 32 h depending on training data size, with training time directly correlated to dataset volume. These computational demands raise questions about trade-offs between larger training datasets and faster model creation times in time-sensitive forensic investigations where rapid response may be critical. Inference performance also varies significantly across deployment approaches. Oh et al. (2024) documented that text generation using local LLM models required approximately 30 min per analysis, with average durations of 10 s for cloud-based ChatGPT versus 3 min for local Llama models. These performance disparities directly impact the practical deployment of LLM-assisted forensic tools in time-sensitive investigative scenarios, potentially creating bottlenecks that negate efficiency gains from automation.

Recent implementations demonstrate varying operational efficiency trade-offs that depend heavily on specific forensic tasks and implementation approaches (Hawa et al., 2025; Shang et al., 2025). Hawa et al. (2025) showed that multimodal LLM analysis reduced average analysis time by 43% while maintaining 88% accuracy compared to expert human analysis, demonstrating substantial efficiency gains for visual evidence processing. However, Shang et al. (2025) found that automated forensic frameworks requiring autonomous interpretation of on-screen information resulted in extended execution times due to sequential processing requirements and action execution delays. These findings suggest that operational efficiency gains may be highly task-dependent, with some forensic applications benefiting substantially from LLM integration, while others may experience performance degradation.

5.6. Domain-specific

Knowledge cutoff date limitations constrain LLM effectiveness in addressing recent and constantly evolving attack vectors (Sreya et al., 2023; Scanlon et al., 2023a). Sreya et al. (2023) noted that LLM understanding is restricted to information available at the time of the most

recent training data update, preventing awareness of recent occurrences, changes, or emerging threats, without access to additional tools such as web search. These temporal limitations prove particularly problematic in dynamic threat environments where forensic investigators must analyse novel or zero-day attack patterns and emerging technologies not represented in historical training data.

LLM proficiency in specialised forensic domains also depends heavily on understanding both technical systems and domain-specific investigative methodologies (Henseler and van Beek, 2023; Cho et al., 2024). Henseler and van Beek (2023) demonstrated that LLM performance degrades substantially when models lack contextual understanding of specific organisational processes, proprietary data formats, or specialised forensic platforms. This limitation reflects the reality that forensic investigations frequently involve unique technical environments and customised systems that may not be well-represented in general training corpora.

Demographic and linguistic specialisation presents additional challenges. Cho et al. (2024) showed that Korean-language LLMs struggled to distinguish between adjacent age groups in author profiling tasks, particularly for the 30s, 40s, and 50s age ranges, possibly due to subtle differences in writing styles that current LLM architectures cannot adequately capture. This finding highlights how domain-specific nuances may exceed current LLM capabilities, requiring specialised training or alternative analytical approaches.

Effective LLM-assisted forensic analysis frequently requires substantial data preprocessing that introduces technical complexity and potential error sources. Kahvedžić (2024) demonstrated that structured preprocessing of chat data improved LLM-based analysis accuracy by 23.4% compared to raw text processing, emphasising the critical importance of domain-specific data preparation in forensic workflows. However, this preprocessing overhead adds additional steps to the investigative process and requires specialised knowledge of both the data format and LLM input requirements. The preprocessing challenge is compounded by the diversity of data formats encountered in forensic investigations. Different evidence sources may require distinct preprocessing approaches, creating complexity in developing generalised LLM-assisted forensic tools. Furthermore, preprocessing decisions may significantly impact analytical outcomes, requiring careful documentation and validation to maintain forensic integrity and support reproducibility requirements.

5.7. Bias and fairness

Systemic bias and fairness concerns in LLM-assisted digital forensics extend beyond technical accuracy issues to encompass fundamental questions of equitable treatment and algorithmic justice. These challenges manifest across training data composition and demographic performance disparities that may stem from such composition, as well as accountability frameworks.

LLMs learn patterns from vast Internet corpora that inherently reflect societal biases and may contain inaccuracies which can propagate into forensic analysis (Scanlon et al., 2023a; Jia et al., 2024). The challenge is compounded by the opacity of training data composition for many commercial LLMs, making it difficult for forensic practitioners to assess potential bias sources or their impacts on specific investigative contexts. Cho et al. (2024) demonstrated that language-specific model training significantly impacts forensic analysis performance, with accuracy degrading to 58.7% when processing mixed-language content. This finding highlights how LLMs may exhibit differential performance across demographic groups, potentially introducing systematic biases into forensic investigations that disproportionately affect certain populations. Such demographic biases raise fundamental fairness concerns, as investigative outcomes may vary based on the linguistic or cultural characteristics of evidence rather than its substantive content.

The ethical framework surrounding LLM use in forensics encompasses transparency and fairness considerations that extend beyond

technical performance metrics (Scanlon et al., 2023b; Wickramasekara and Scanlon, 2024). Scanlon et al. (2023b) raised essential questions about the appropriate level of investigator reliance on LLM outputs and the mechanisms needed to ensure fair and non-maleficent application of these technologies. The black-box nature of LLM decision-making processes presents particular challenges for maintaining transparency in forensic investigations, where the ability to explain analytical methods is crucial for both legal proceedings and public accountability. When automated systems contribute to investigative decisions that may affect legal outcomes, determining accountability for errors becomes complex. Should responsibility rest with the forensic practitioner who employed the LLM, the organisation that deployed the system, or the model provider whose training decisions influenced outputs? Wickramasekara and Scanlon (2024) acknowledged risks including information accuracy and legal barriers in their proposed multi-agent framework; however comprehensive governance frameworks addressing these accountability questions remain underdeveloped.

The informal nature of chat data and conversational evidence presents additional ethical challenges related to interpretation and potential mischaracterisation (Kahvedžić, 2024). LLMs trained on casual communication patterns may not adequately handle the nuanced requirements of forensic contexts where informal language, slang, sarcasm, and cultural references require careful interpretation. Misinterpretation of informal communication could lead to incorrect conclusions that unfairly characterise individuals or misrepresent their intentions, raising concerns about the fairness and accuracy of LLM-assisted analysis in contexts involving everyday digital communications.

The absence of established ethical guidelines for LLM use in forensic contexts creates uncertainty for practitioners and organisations seeking to deploy these technologies responsibly. Whilst technical capabilities continue to advance, the development of corresponding ethical frameworks, professional standards, and governance mechanisms lags substantially behind, creating a gap that may expose both forensic organisations and investigated individuals to potential harm from inappropriate or inadequately validated LLM applications.

5.8. Summary of challenges and limitations

These multifaceted challenges collectively demonstrate that whilst LLMs offer significant potential for enhancing digital forensic capabilities, substantial research and development efforts remain necessary before these technologies can achieve widespread operational deployment whilst maintaining the scientific rigour and legal defensibility critical to digital forensic science. The path forward requires coordinated efforts across several domains, with particular emphasis on developing forensic-specific solutions, rather than attempting to adapt general-purpose LLMs to forensic requirements through basic prompt engineering alone.

6. Future research directions

The comprehensive analysis of challenges and limitations presented in Section 5 establishes a foundation for identifying critical research priorities for the advancement of LLM-assisted digital forensics towards operational maturity. As illustrated in Table 3, these research directions span several domains, with each addressing multiple challenge categories identified throughout this survey. These priorities build upon the three strategic integration points established in Section 4, focusing on advancing LLM capabilities in pattern recognition, evidence analysis, and evidence presentation whilst addressing the fundamental challenges that currently constrain practical operational deployments.

6.1. Technical research needs

6.1.1. Forensic-ready LLM architectures

The development of forensic-ready LLM architectures requires fundamental architectural innovations that embed forensic requirements

directly into model design, rather than attempting to retrofit general-purpose models for forensic applications. Built-in logging and audit capabilities must capture comprehensive traces of model invocations, including input prompts, generated outputs, intermediate reasoning steps, and all relevant parameters, creating immutable audit trails that support chain-of-custody requirements and allow for retrospective validation of LLM-assisted analytical processes. Minimally probabilistic operation modes, specifically designed for forensic contexts should support

reproducible outputs across multiple invocations of identical inputs, potentially through controlled sampling mechanisms or temperature-zero configurations that minimise probabilistic variation. Explainability-by-design approaches must move beyond post-hoc interpretation techniques to incorporate transparent reasoning mechanisms that generate human-interpretable justifications for analytical conclusions, addressing the black-box nature of current LLM architectures that fundamentally conflicts with forensic transparency requirements.

Table 3

Mapping of future research directions to identified challenges in LLM-assisted digital forensics. Research directions are organised by domain and mapped to the primary challenge categories they aim to address.

Research domain	Future research direction	Technical	Methodological	Forensic integrity	Data privacy	Operational	Domain-specific	Bias and fairness
<i>Technical research needs</i>								
Forensic-ready architectures	Built-in logging and audit capabilities	•	◦	•	◦	•	◦	◦
	Operation modes for forensic contexts	•	◦	•	◦	◦	◦	◦
	Explainability-by-design approaches	•	◦	•	◦	◦	◦	•
Advanced analysis techniques	Multi-model ensemble approaches for reliability	•	•	◦	◦	•	◦	◦
	Uncertainty quantification methods	•	•	•	◦	◦	◦	◦
	Adversarial robustness improvements	•	◦	•	•	◦	◦	◦
Specialised forensic LLMs	Real-time forensic monitoring capabilities	•	◦	◦	◦	•	•	◦
	Domain-specific fine-tuning methodologies	•	•	◦	◦	◦	•	◦
	Privacy-preserving local deployment architectures	◦	◦	•	•	•	◦	◦
	Forensic-optimised prompt engineering frameworks	◦	•	◦	◦	•	•	◦
	Purpose-built forensic reasoning models	•	◦	◦	◦	◦	•	◦
<i>Methodological research needs</i>								
Validation frameworks	Standardised evaluation protocols	◦	•	•	◦	•	◦	◦
	Benchmark dataset development	◦	•	◦	◦	◦	•	◦
	Forensic-specific performance metrics	◦	•	•	◦	◦	◦	◦
	Reliability assessment methodologies	•	•	•	◦	◦	◦	◦
Process model refinement	Empirical validation of DFRWS extensions	◦	•	•	◦	•	◦	◦
	Best practice guidelines development	◦	•	•	◦	•	◦	•
	Quality assurance protocols	◦	•	•	◦	•	◦	◦
Integration strategies	Hybrid human-AI workflow design	◦	•	•	◦	•	◦	•
	Tool ecosystem development	◦	•	◦	◦	•	•	◦
	Training curriculum design	◦	•	◦	◦	•	◦	◦
	Organisational adoption roadmaps	◦	•	◦	◦	•	◦	◦
<i>Legal and policy research needs</i>								
Legal framework development	Admissibility standards for LLM-assisted evidence	◦	◦	•	◦	◦	◦	◦
	Liability and accountability frameworks	◦	◦	•	◦	◦	◦	•
	Cross-jurisdiction harmonisation	◦	◦	•	•	◦	◦	◦
Ethical guidelines	Regulatory guidance development	◦	◦	•	•	•	◦	•
	Bias detection and mitigation strategies	◦	•	◦	◦	◦	◦	•
	Fairness assessment frameworks	◦	•	•	◦	◦	◦	•
	Transparency requirements	◦	◦	•	◦	◦	◦	•
<i>Cross-cutting research priorities</i>								
Interdisciplinary collaboration	AI-forensics-law research partnerships	◦	•	•	◦	◦	◦	•
	Industry-academia knowledge transfer	◦	•	◦	◦	•	•	◦
	International research consortia	◦	•	•	•	◦	◦	◦
Empirical validation	Real-world deployment studies	◦	•	•	◦	•	•	◦
	Longitudinal performance assessment	•	•	◦	◦	◦	◦	◦
	Comparative effectiveness research	◦	•	◦	◦	•	◦	◦

6.1.2. Advanced analysis techniques

Advanced analysis techniques must address the reliability, quality, and robustness challenges inherent to single-model LLM deployments through systematic integration of complementary approaches. Multi-model ensemble architectures and multi-agent orchestration workflows that combine outputs from diverse LLM architectures can potentially mitigate individual model limitations by reducing hallucination risks and improving output quality through various consensus mechanisms, though such approaches require careful validation to ensure that ensemble predictions maintain forensic integrity and do not simply amplify shared biases across the individual models. Uncertainty quantification methods represent another potential critical research priority, enabling LLMs to provide calibrated confidence estimates alongside analytical outputs, so that forensic practitioners can prioritise human verification efforts based on model uncertainty. Furthermore, adversarial robustness improvements must address the susceptibility of LLMs to manipulation through carefully crafted inputs via prompt injection attacks, particularly relevant given the adversarial nature of criminal investigations where sophisticated actors may employ sophisticated anti-forensic techniques.

6.1.3. Specialised forensic LLMs

The development of specialised forensic LLMs represents a critical research direction that moves beyond general-purpose model adaptation towards purpose-built architectures and knowledge sets optimised for forensic requirements. Domain-specific fine-tuning methodologies must extend beyond simple instruction tuning to incorporate forensic-specific reasoning patterns, legal reasoning rules and terminology, and investigative workflows, building upon foundational work such as ForensicLLM (Sharma et al., 2025). These specialised models should demonstrate improved output quality and reduced hallucination rates compared to general-purpose alternatives when handling digital forensic artefacts. Furthermore, privacy-preserving local deployment architectures require investigation into efficient model compression techniques, quantisation strategies, and hardware acceleration approaches that enable the deployment of capable forensic LLMs within resource-constrained environments whilst maintaining data confidentiality requirements and acceptable performance levels. Also, forensic-optimised prompt engineering frameworks should codify best practices for eliciting reliable and high-quality analytical outputs from LLMs across diverse forensic contexts and tasks, potentially incorporating structured prompt templates and validation checkpoints to reduce sensitivity to prompt variations and improve consistency across different scenarios.

6.2. Methodological research needs

6.2.1. Validation frameworks

Standardised evaluation protocols must establish consistent methodologies for assessing LLM performance across diverse digital forensic contexts, addressing the current fragmentation where different studies employ a wide variety of metrics and evaluation criteria that prevent meaningful cross-study comparisons. The associated benchmark dataset development requires the creation of comprehensive, publicly available forensic datasets that capture the complexity and diversity of real-world investigations. Forensic-specific performance metrics could extend beyond traditional accuracy measures to encompass reliability, consistency, explainability, and legal admissibility considerations, building upon initial efforts such as Evidence Extraction Accuracy (EEA) and Forensic Artefact Precision (FAP) (Khatriwala et al., 2025) to develop comprehensive evaluation frameworks that align with forensic integrity requirements. Reliability assessment methodologies should provide systematic approaches for quantifying LLM consistency across multiple invocations, thereby evaluating performance degradation under adverse conditions empirically, and establishing confidence bounds that support risk-based validation strategies in operational forensic contexts.

6.2.2. Process model refinement

Empirical validation of the proposed DFRWS extensions must move beyond theoretical frameworks to demonstrate practical effectiveness and real utility through controlled experiments and real-world deployment studies, addressing the current gap where comprehensive integration frameworks remain largely unvalidated in operational forensic contexts. Best practice guidelines development requires systematic codification of lessons learned from early LLM deployments in forensic contexts, establishing evidence-based recommendations for LLM and parameter selection, prompt engineering, validation protocols, and human oversight mechanisms that balance automation benefits with forensic integrity requirements. Quality assurance protocols must also define systematic approaches for validating LLM outputs to establish appropriate thresholds for human verification, and documenting LLM-assisted forensic processes in ways that support legal admissibility and scientific reproducibility. These protocols should explicitly address the technical quality challenges of hallucination detection and output consistency verification.

6.2.3. Integration strategies

Hybrid human-AI workflow design must establish the optimal division of labour between LLM capabilities and human expertise, enumerating specific forensic tasks where automation and AI-based enhancement provide measurable benefits whilst maintaining required quality standards. The development of the associated tool ecosystem requires the creation of interoperable forensic platforms that can incorporate LLM capabilities alongside traditional forensic tools to create combined unified workflows with built-in quality assurance mechanisms. Training curriculum design must address the skill gaps identified across the literature, developing educational programmes that equip forensic practitioners with LLM knowledge and prompt engineering expertise, LLM validation capabilities including output quality assessment, and critical evaluation skills necessary for the effective deployment of AI-assisted forensic workflows. Finally, organisational adoption roadmaps should provide structured guidance covering technical infrastructure requirements, policy development and training needs, and phased deployment strategies to manage risks as part of incremental capability development whilst establishing quality benchmarks at each deployment phase.

6.3. Legal and policy research needs

6.3.1. Legal framework development

Admissibility standards for LLM-assisted evidence must be established through systematic engagement with legal communities, addressing how AI-generated analytical outputs can meet established criteria such as the Daubert standard. Liability and accountability frameworks require clarification of responsibility allocation when LLM-assisted analysis contributes to investigative decisions and defining appropriate standards of care for AI-assisted forensic practice. Cross-jurisdiction harmonisation efforts must address the reality that digital investigations frequently span multiple legal systems with varying standards for evidence admissibility, data privacy requirements, and AI governance frameworks, requiring international coordination to establish consistent approaches that support effective cross-border investigations. Finally, regulatory guidance development should provide clear direction for forensic organisations regarding appropriate LLM deployment, validation requirements, documentation standards, and oversight mechanisms, reducing uncertainty that currently constrains operational adoption of these technologies in real-life investigative contexts.

6.3.2. Ethical guidelines

Bias detection and mitigation strategies must address the reality that LLMs trained on Internet corpora inherently reflect societal biases that can propagate into forensic analysis. These strategies operate at two levels – first, addressing technical accuracy degradation and output quality issues caused by biased training data (as discussed in the technical challenges of Section 5), and second, ensuring systemic fairness

by identifying demographic disparities in model performance and implementing corrective measures that ensure equitable treatment across diverse populations. This dual approach requires systematic methodologies for both technical validation of output quality across different input types and fairness assessment of investigative outcomes across demographic groups, linguistic communities, and cultural contexts.

Fairness assessment frameworks should establish methodologies for evaluating whether LLM-assisted forensic analysis produces equitable outcomes across different demographic groups, addressing concerns that automated systems may systematically disadvantage certain populations through either technical quality degradation or inherent systemic bias. The associated transparency requirements must define appropriate levels of disclosure regarding LLM deployment in forensic investigations.

6.4. Cross-cutting research priorities

Interdisciplinary collaboration represents a fundamental requirement for advancing LLM-assisted digital forensics, necessitating sustained partnerships between AI researchers, forensic practitioners, and legal scholars that bridge disciplinary boundaries. Relevant industry-academia knowledge transfer mechanisms can facilitate bidirectional exchange of insights, with academic researchers gaining access to real-world forensic challenges and operational constraints, while practitioners benefit from cutting-edge technical developments and rigorous scientific evaluation methodologies.

Empirical validation through real-world deployment studies remains critically important, moving beyond controlled laboratory experiments to assess LLM performance in operational forensic environments where data heterogeneity and adversarial conditions may significantly impact effectiveness. Longitudinal performance assessment could track LLM capabilities over extended periods, evaluating how model updates and evolving deployment approaches affect forensic utility over time. Comparative effectiveness research must therefore systematically evaluate alternative LLM architectures, deployment strategies, and integration approaches across diverse digital forensic contexts, establishing evidence-based guidance for practitioners regarding optimal technology selection and implementation approaches that balance operational performance, cost, privacy, and forensic integrity requirements.

7. Discussion

7.1. Key findings summary

The presented systematic review describes the rapid progression of LLM integration in digital forensics from exploratory investigations in 2023 to empirical validation efforts in 2025, with research predominantly concentrated in the examination and analysis phases of the DFRWS process model. The literature establishes that LLMs achieve substantial performance improvements across diverse forensic tasks, including detection accuracies exceeding 85% in threat hunting (Lin, 2024), 94.6% precision in log anomaly detection (Pan et al., 2024), and 98% classification accuracy in specialised evidence extraction (Kim et al., 2025a). However, critical challenges persist across a number of key areas. The fundamental misalignment between the probabilistic nature of LLM architectures and the deterministic requirements of forensic science remains the central challenge constraining operational deployment.

7.2. Implications for practice

Forensic practitioners should approach LLM integration as an augmentation strategy for human expertise, prioritising applications in evidence discovery, pattern recognition, and preliminary analysis where LLM capabilities align with investigative requirements combined with human oversight. Technical approach selection must balance performance capabilities against deployment constraints, with cloud-based frontier models offering superior capabilities at the cost of

data privacy concerns, whereas local open-weight alternatives such as ForensicLLM (Sharma et al., 2025) provide privacy-preserving options with potentially reduced performance and increased infrastructure requirements. Alongside practical deployment, the associated risk mitigation strategies should incorporate comprehensive documentation of LLM invocations, parameters, and outputs to support chain-of-custody requirements, and establish clear governance frameworks defining appropriate use cases, oversight mechanisms, and accountability structures for LLM-assisted forensic analysis.

7.3. Implications for research

This survey establishes a preliminary framework for systematically evaluating LLM integration in digital forensics through two complementary dimensions. The *capability* dimension, mapped to the DFRWS process model phases, provides a structured approach for assessing where LLMs can provide measurable benefits across digital forensic workflows. As illustrated in Fig. 5, this dimension identifies three strategic integration points – (1) pattern recognition during examination, (2) evidence analysis, and (3) evidence presentation. The *challenge* dimension, presented in Fig. 8 and detailed in Section 5, establishes seven analytical lenses through which researchers can examine constraints and limitations in LLM-assisted digital forensics. Together, these dimensions provide a systematic framework for evaluating future LLM-assisted forensic approaches.

Priority research areas presented in Section 6 must address the fundamental challenges of explainability, reproducibility, and legal admissibility through the development of forensic-ready LLM architectures that embed audit capabilities, transparent reasoning mechanisms, and predictable operation modes directly into model design rather than attempting to retrofit general-purpose models for forensic applications. Interdisciplinary collaboration between AI researchers, forensic practitioners, and legal scholars represents a critical enabler for advancing this field, requiring sustained partnerships that facilitate bidirectional knowledge transfer and the establishment of standardised evaluation frameworks. The long-term research agenda must prioritise empirical validation using real-world forensic data instead of extensive reliance on synthetic datasets, combined with a systematic investigation of adversarial robustness against anti-forensic techniques.

7.4. Limitations of current survey

This survey focuses specifically and exclusively on LLM applications in digital forensics, excluding broader AI techniques such as computer vision, traditional ML, and specialised neural architectures that also contribute to forensic capabilities, thereby limiting the scope to natural language processing and reasoning capabilities inherent to LLMs at the time of writing. The rapidly evolving landscape of LLM capabilities means that findings may become outdated as new models, architectures, and deployment approaches emerge, with the literature review capturing works published through October 2025 and potentially missing subsequent developments in this highly dynamic field.

8. Conclusion

We presented a comprehensive examination of the emerging intersection between generative AI and digital forensics, synthesising 33 peer-reviewed works mapped across the phases of the DFRWS process model. Our analysis demonstrates that LLMs can achieve substantial performance improvements across diverse forensic tasks, yet their inherently probabilistic nature is not congruent with the deterministic requirements of digital forensic science. The path forward requires coordinated interdisciplinary research bridging AI, digital forensics, and law to devise forensic-ready LLM architectures that embed explainability, reproducibility, and legal admissibility requirements, alongside standardised validation frameworks and comprehensive governance mechanisms. As the field progresses from exploratory investigations towards operational maturity across specialised forensic tasks, the researcher

and practitioner community must prioritise human-AI collaborative approaches that can leverage LLM capabilities for pattern recognition, preliminary analysis and evidence presentation alongside rigorous human oversight and strict validation controls.

CRedit authorship contribution statement

Maxim Chernyshev: Writing – original draft, Methodology, Conceptualization. **Zubair Baig:** Writing – review & editing, Supervision, Conceptualization. **Naeem Syed:** Writing – review & editing. **Robin Doss:** Writing – review & editing. **Malcolm Shore:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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